



INTEGRATED SYSTEMS ARCHITECTURE

POLITECNICO DI TORINO

DIPARTIMENTO DI ELETTRONICA E TELECOMUNICAZIONI

design and implementation of the IIR filter

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Contents

1	Reference model development	3
1.1	Filter design and coefficient quantization using Matlab	3
1.2	Testing the filter and fixed point implementation	4
1.2.1	First step: Matlab pseudo-fixed-point	4
1.2.2	Second step: fixed-point C model and calculation of the THD	6
1.2.3	Explanations,Comparitions and comments	6
2	VLSI implementation	7
2.1	Architecture	7
2.2	Architecture Development in VHDL and Simulation in QSim	8
2.3	Logic synthesis	9
2.3.1	Without clock gating	10
2.3.2	With clock gating	12
2.3.3	Place and Route	13
3	Advanced architecture development	16
3.1	Filter design and coefficient quantization	16
3.1.1	Application of J-Lookahead Approach	16
3.1.2	pipelining and retiming	18
3.1.3	final design	18
3.2	Testing the filter and fixed point implementation	18
3.3	VLSI implementation	19
3.3.1	architecture explanation	19
3.4	VHDL and Simulation	19
3.4.1	Simulation Verification in Qsim	21
3.5	Logic Synthesis	21
3.5.1	Without Clock Gating	21
3.5.2	With Clock Gating	23
3.5.3	Comparison of Results	24
3.6	place and route	25
4	Appendix	27
4.1	myfilterii.c	27
4.2	sample.txt	29
4.3	resultc_17.txt	32
4.4	thd filter	35
4.5	fir.vhd	36
4.6	ad_comand_synthesis.txt	38
4.7	area_without_clk_gate.txt	39
4.8	timing_without_clk_gate.txt	40
4.9	comand_power_estimation.txt	42
4.10	power_withoutCLK_GATE.txt	43
4.11	area_clk_gate.txt	44
4.12	timing_clk_gate.txt	45
4.13	power_CLK_GATE.txt	47

4.14	timing_report_innovus.txt	49
4.15	area_report_innovus.txt	52
4.16	myfiir_advanced.c	53
4.17	resultc_18_lookahead.txt	55
4.18	ad_area_without_clk_gate.txt	58
4.19	ad_power_without_clk_gate.txt	59
4.20	ad_timing_without_clk_gate.txt	60
4.21	ad_area_clk_gate.txt	61
4.22	ad_power_clk_gate.txt	63
4.23	ad_timing_clk_gate.txt	64
4.24	fiir-lookahead.vhd	67

1 Reference model development

The goal of this lab is to design and implement a digital filter that has a cut-off frequency of 2 kHz, with a sampling frequency set at 10 kHz. Depending on our group number(number 5),the parameter p is set to 0 and hence we need to design an Infinite Impulse Response (IIR) filter. Figure 1a and figure 1b gave us general idea of how to design the filter and test it.

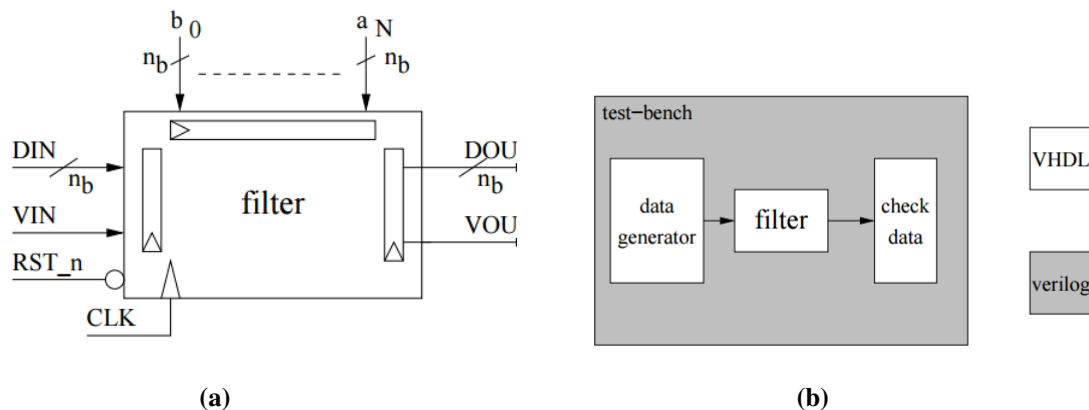


Figure 1: Schematic of Filter

1.1 Filter design and coefficient quantization using Matlab

Using the butter function, we calculate the quantized coefficients. Two key parameters for the design of filter are the filter order N and the number of bits for the coefficients nb . These parameters were determined using the following formulas:

$$N = 2^p \cdot [(x \text{ mode } 2) + 1] + 6 \cdot p \quad (1)$$

$$N = 1$$

$$nb = (y \text{ mod } 7) + 8 \quad (2)$$

$$nb = 13$$

$$x = 1, y = 13$$

x and y are determined based on the number our surnames characters: XU SHANG

So the number of coefficients is equal to $N + 1 = 2$ and each coefficient would be represented in $nb = 13$ bits.

1.2 Testing the filter and fixed point implementation

1.2.1 First step: Matlab pseudo-fixed-point

The primary aim of this step is to develop and test a pseudo-fixed-point IIR filter using MATLAB, verifying its effectiveness in passing specific signal frequencies and attenuating others while simulating fixed-point limitations.

In this step, using the provided MATLAB files, we designed a digital IIR filter and quantized its coefficients to 13 bits to approximate fixed-point behavior, 1 bit as signed integer bit and 12 bits for fractional part.

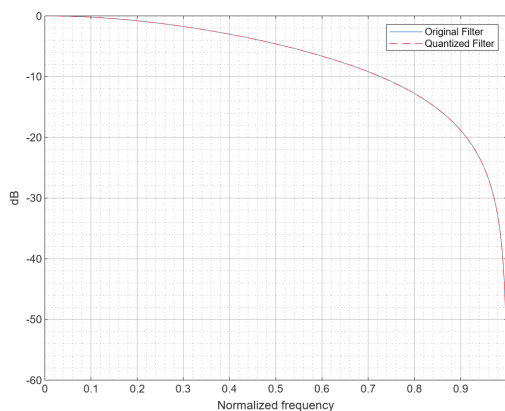
Obtained quantized coefficients values :

quantized coefficient	decimal value	binary value
a0	1.000000000000000	0.111111111111
a1	-0.158447265625000	1.110101110111
b0	0.420654296875000	0.011010111011
b1	0.420654296875000	0.011010111011

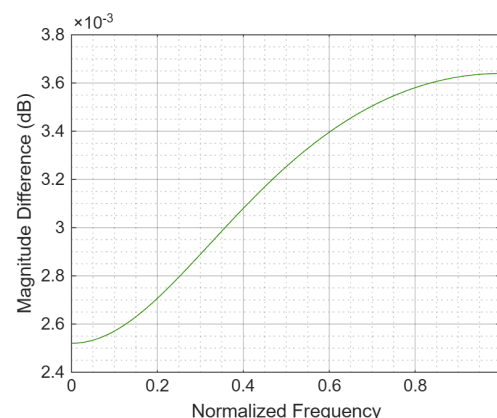
Table 1

$$b_i = 1723 \quad 1723$$

$$a_i = 4096 \quad -649$$



(a) Frequency Response



(b) Magnitude Deviation, Real vs. Quantized case

Figure 2

The figure2a depicts that generally, by the quantization of coefficients in 13 bits we get frequency response close to the case with fixed point coefficients. However, by looking more precisely at figure 2b we can see that as frequency increases the errors become more pronounced. which shows at high frequencies, where the rapid oscillations depend heavily on precise cancellations, small movement of Zeros and Poles due to quantization error can lead to a noticeable change in the filter's frequency response. which in this case the magnitude of deviation in frequency response reach to maximum value of $3.6 \times 10^{-3} dB$ at frequency equal to Nyquist frequency.

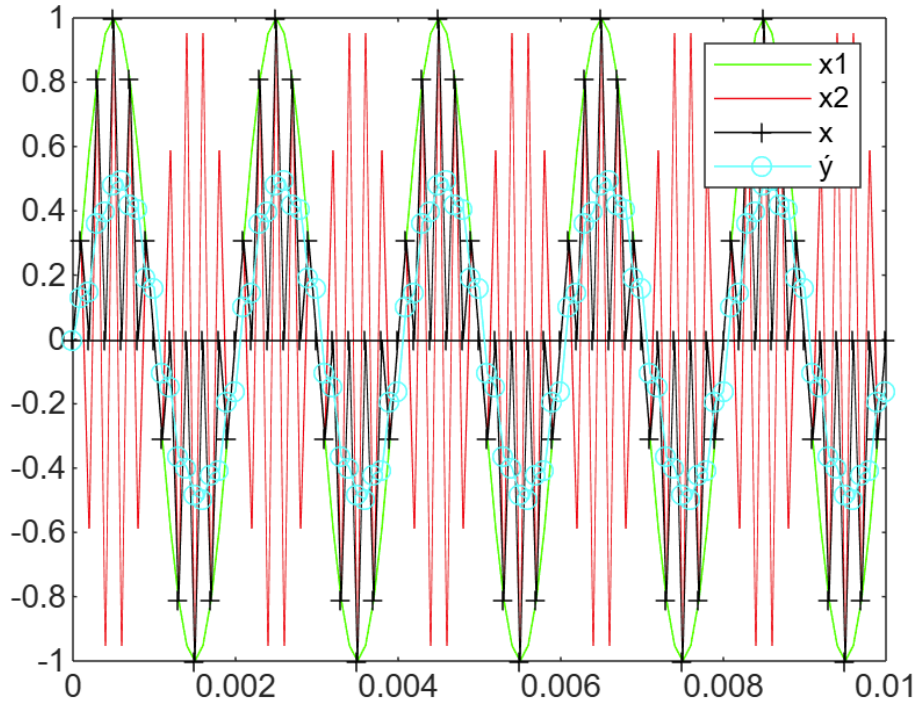


Figure 3: Time-Domain Representation of Input and Filtered Output

In addition to that, this quantized filter was tested in MATLAB by applying it to a composite input signal x , composed of two frequency components (500 Hz and 4500 Hz), allowing us to observe how effectively the filter attenuates the out-of-band frequency (4500 Hz) while passing the in-band frequency (500 Hz).

The plot in Figure 3 illustrates the filter's selectivity, showing that it largely retains the 500 Hz component while significantly reducing the 4500 Hz component.

Finally, to further simulate fixed-point representation, the input and output signals were quantized to 13 bits. The quantized input (x_q) and output (y_q) have been saved into files `resultsm.txt` and `samples.txt` for further analysis.

Workspace	
Name	Value
f	1x201 double
fft_resultc	201x1 complex d...
fft_resultm	201x1 complex d...
fs	10000
magnitude_c	201x1 double
magnitude_m	201x1 double
n	201
resultc_data	201x1 double
resultm_data	201x1 double
THD_filter_ll_c	-41.9217

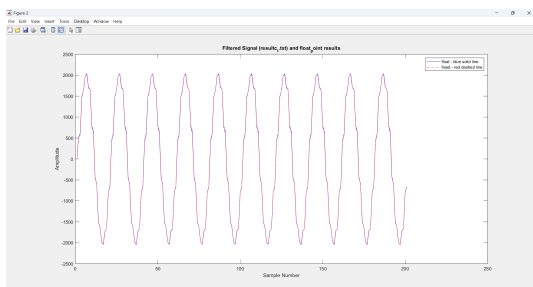
Figure 4: Result of THD with shmat=17

1.2.2 Second step: fixed-point C model and calculation of the THD

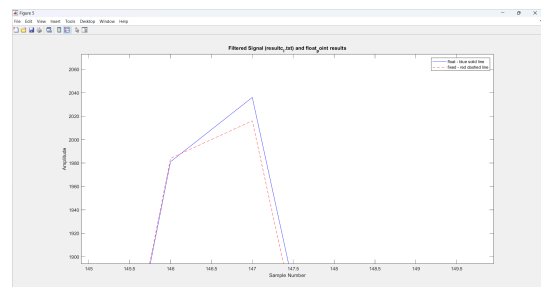
In this section, we are tasked with writing a C program that uses the provided `samples.txt` file (downloaded from the specified website) as input data and generates an output file called `resultc.txt`. Subsequently, we add a piece of data in MATLAB code and use the `thd` function to evaluate the Total Harmonic Distortion (THD) of the result y from our fixed-point implementation. After performing multiplications, we decide to discard 17 bits, achieving a THD of -41.9217 dB (as shown in Fig. 4), which closely meets the specified requirements. The parameter $SHAMT = 17$, used in the C code, represents the shift amount. The C program and MATLAB code are provided in Appendix 4.1 and Appendix 4.4, respectively. Due to an overflow issue. The output file, named `resultc_17` ($SHAMT = 17$), is included in Appendix 4.3.

1.2.3 Explanations,Comparitions and comments

As we can see in Figure 5a,we compare two different result file we got from C and Matlab,there almost little difference between them, what if we zoom in on one of the sections, we can see that the peak difference between the two curves is 0.97, which corresponds to -0.2 dB(Fig: 5b); the level of attenuation is acceptable.



(a) result of the filter in float method and fixed method



(b) zoom float and fix.

Figure 5

2 VLSI implementation

In this chapter we will describe the results we obtained after the synthesis phase, in particular regarding the area, the timing and the power consumption of our Infinite Impulsive Response Filter.

2.1 Architecture

The architecture of the filter, that we have seen before, is shown in Figure 6

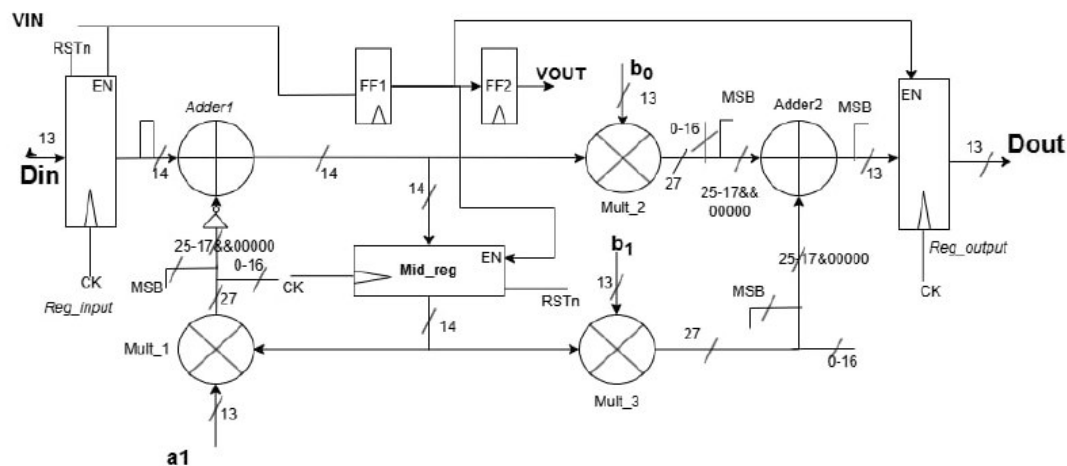


Figure 6: IIR filter schematic

This architecture includes a series of registers, adders, and multipliers. We decide to switch to an internal 14-bits instead of 13-bit because of the overflow.

Also, there are two important flip-flops, FF1 and FF2, further enhances stability by buffering intermediary data, which reduces critical path delay and mitigates setup time violations in high-speed operations. The enable signals control when the data is valid at the output VOUT, ensuring synchronous data transfer and correct signal timing for subsequent processing stages.

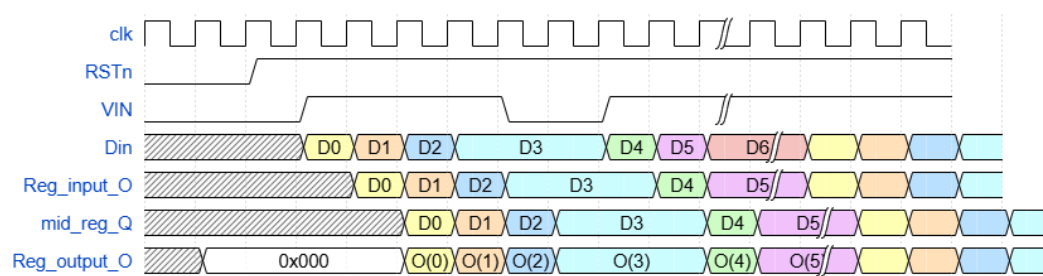


Figure 7: Timing diagram

The Figure 7 shows that each input data requires two clock cycles in order to be processed at the output, due to the existence of two registers on any full path in the circuit.

2.2 Architecture Development in VHDL and Simulation in QSim

Based on the previous architecture, implemented in VHDL, focusing on data flow and control logic and the structural representation, (vhdl code is included in appendix 4.5).

Using the test bench provided in material, we obtained an output file containing the results generated by our I.I.R filter. These results are identical to those produced by the *myfilterii.c* program. In Qsim, when simulating the entire architecture using *data_sinker.vhd*, if the results generated by *fiir.vhd* are not equivalent to those generated by the C program, a note highlighting the differences is provided for comparison. However, as shown in Fig. 8, the results from both implementations are identical.

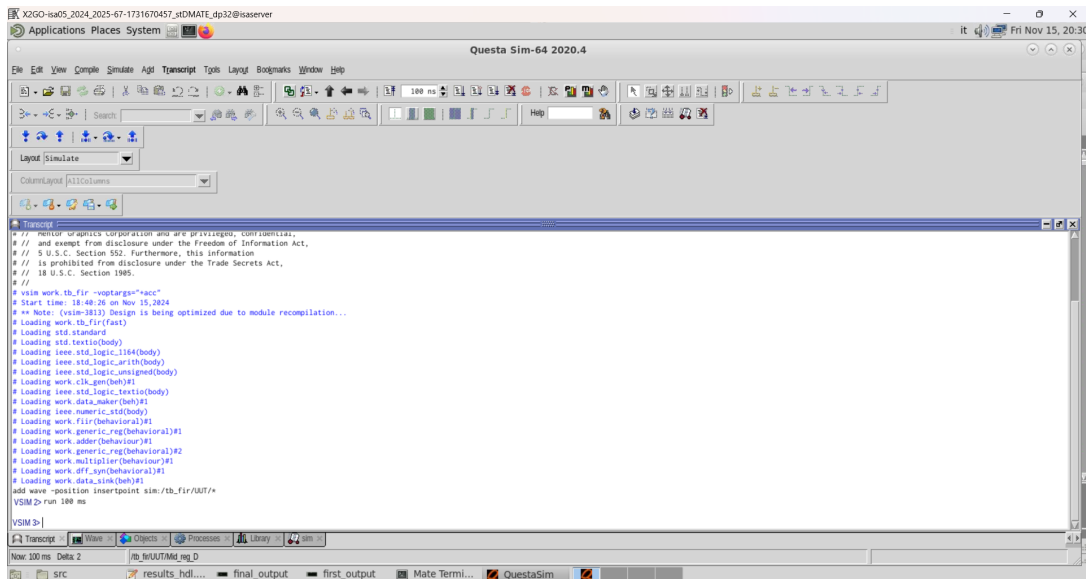


Figure 8: screenshot of Qsim

As shown in 9, The VIN signal toggles from '0' to '1', remains at '1' for several clock periods, then transitions to '0' for one clock period before toggling back to '1', the results are still identical to which generated by C program. The reason is when the Vin toggles to low level, the sampled input will be invalid, the following output in the next clock cycle also will not be considered.

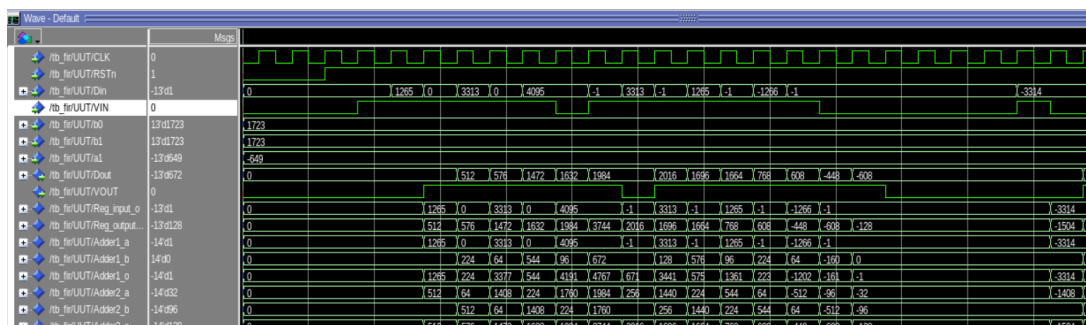


Figure 9: screenshot of waveform

As specified in the *es1v3.0 description.pdf*, we set the clock period to 100,000 ns (f=10 kHz). The total time required to complete the simulation of this entire process—considering the

duration from the first rising edge of the clock when sampling $V_{out} = '1'$ to the last rising edge of the clock when sampling $V_{out} = '1'$ —is 38,800,000 ns, or 38.8 ms(as show in Figure 10 and Figure 11.

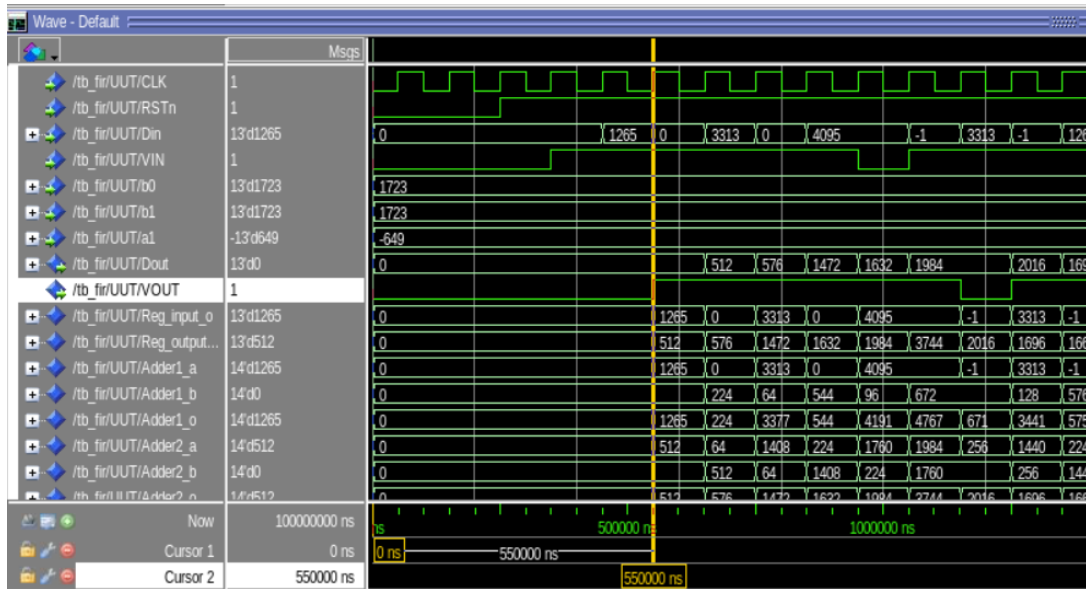


Figure 10: waveform: first high level Vout

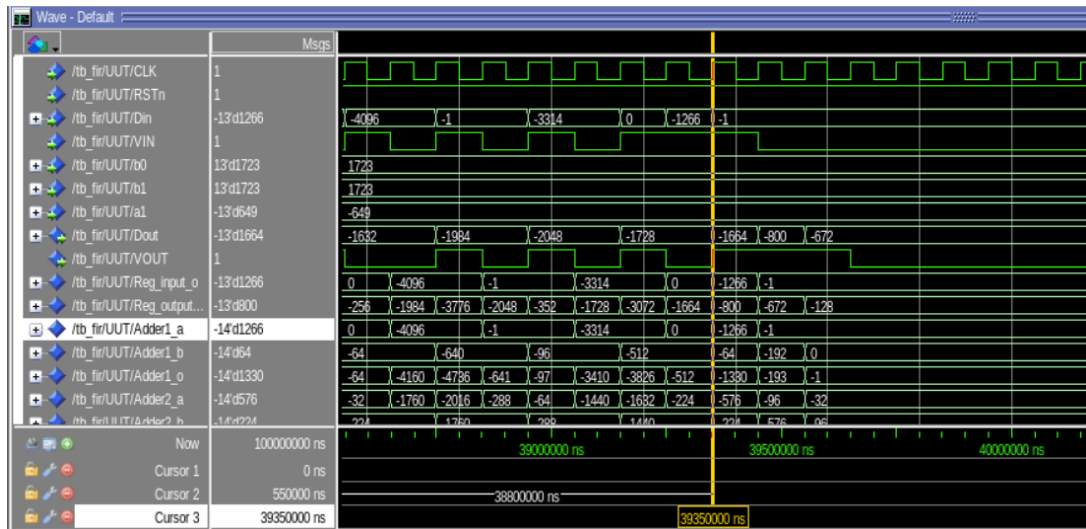


Figure 11: waveform: last high level Vout

By using schematic viewer, we obtain the top level schematic of the testbench in Fig 12, which is matching to the Fig 1.b in *es1v3.0 description.pdf*. The Register Transfer Level diagram of the filter is shown in Fig 13

2.3 Logic synthesis

In this step, we import our VHDL file into the Synopsys Design Compiler to generate an optimized architecture for the IIR filter. The synthesizer is able to generate area and timing

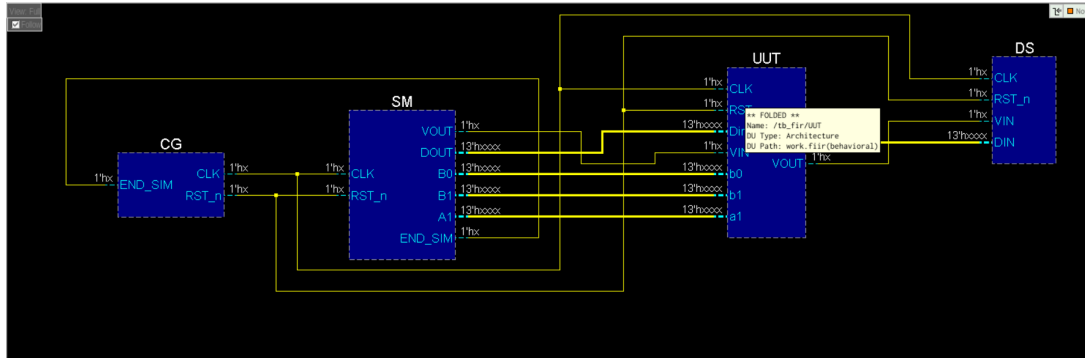


Figure 12: schematic: testbench

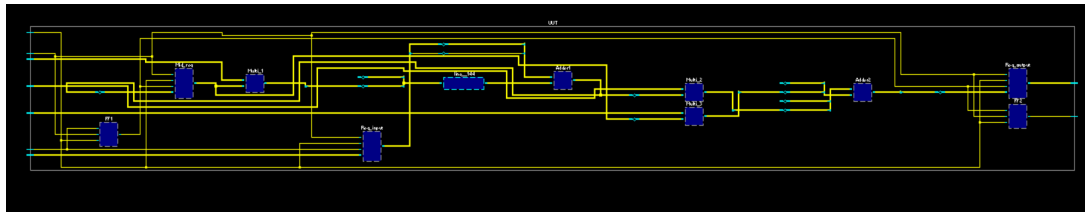


Figure 13: schematic of UUT filter

reports, which provide the area details of the final architecture and the path delay.

2.3.1 Without clock gating

synthesis and simulation

Using the command shown in comand-synthesis.txt, we compile the new architecture without clock gating technology. The maximum clock frequency at which the design can operate correctly is 180.8318 MHz (corresponding to a delay of 5.53 ns, as shown in Appendix 4.12). The corresponding area report, included in Appendix 4.7, indicates that the total area is 2591.372015 μm^2 .

Finally, we generate a new fir.v file under the netlist folder and verify it by testing at the previously determined maximum clock frequency; the result is as same as the results generated in c program, and as shown in the Fig 14 and Fig15, the time distance between the first valid output and the last valid output is 2340 ns.

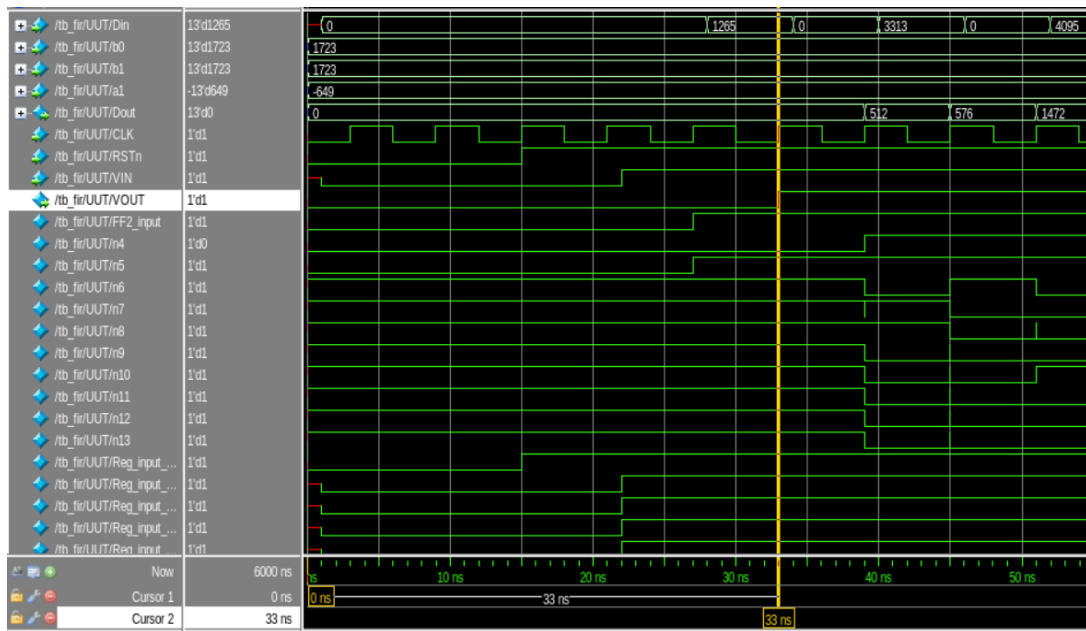


Figure 14: waveform: first high level Vout

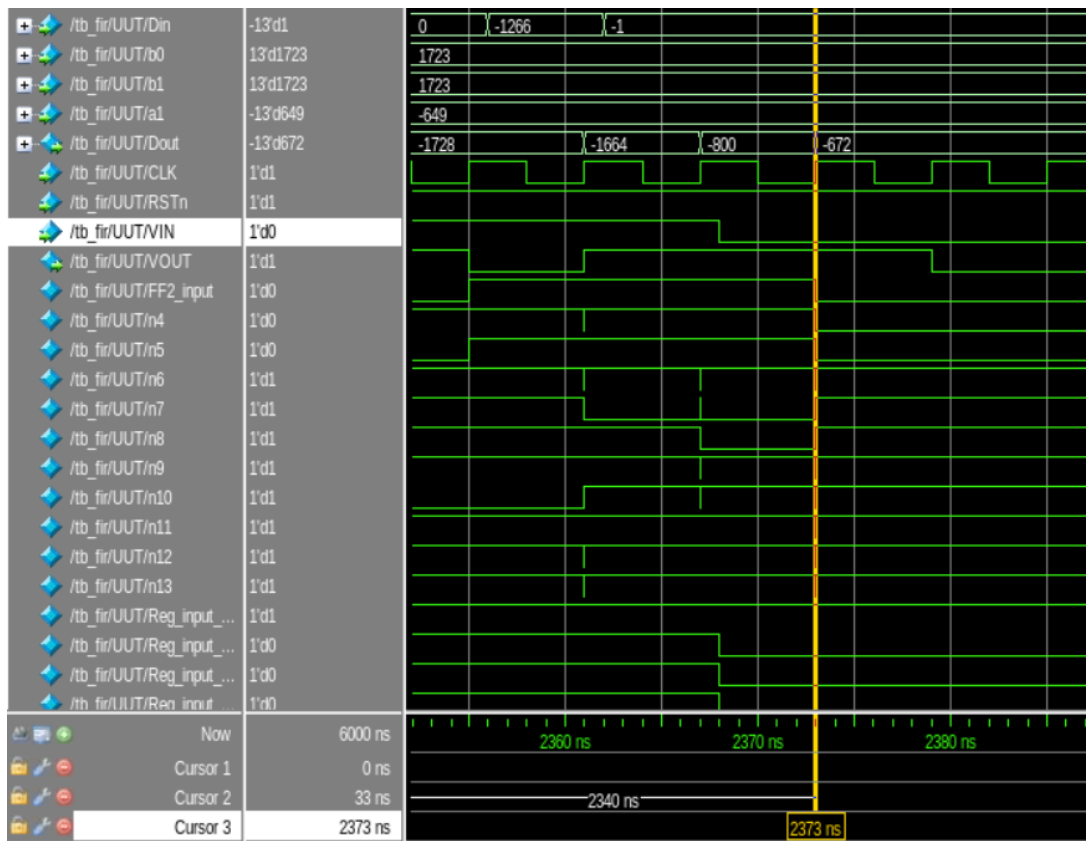


Figure 15: waveform: last high level Vout

power estimation

Using comand as shown in Appedix4.9, we got the power estimation report in Appendix 4.10, the total power estimation is 389.0963 uW.

2.3.2 With clock gating

We repeated the process after synthesizing with the clock gating option enabled and observed slightly different results. With clock gating enabled, the minimum period improved to 5.48 ns (Appendix 4.12), allowing a maximum frequency of 182.481752 MHz. At this frequency, the cell area measured was 2498.006017 μm^2 (Appendix 4.11), and the power consumption was recorded as 234.9579 μW (detailed reports are available in Appendix 4.13).

As same as previous case, time distance between the first valid output(26 ns as shown in Figure 16) and the last valid output(1586 ns as shown in Figure 17) is 1560 ns.

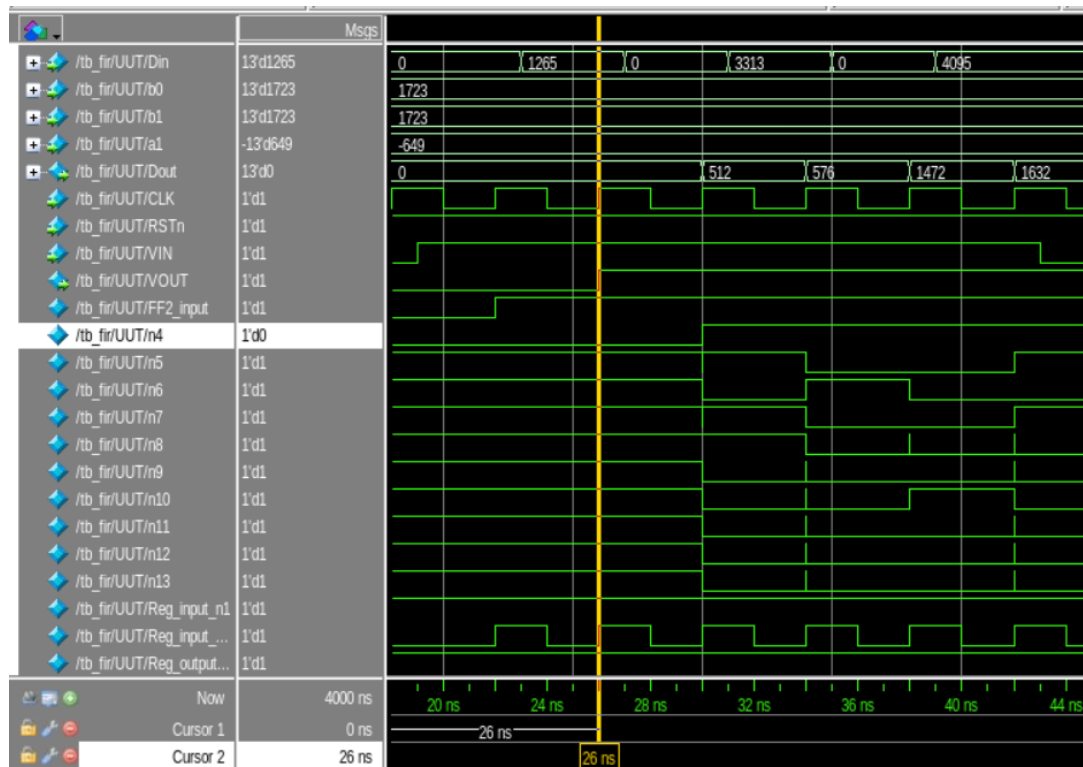


Figure 16: waveform: first high level Vout

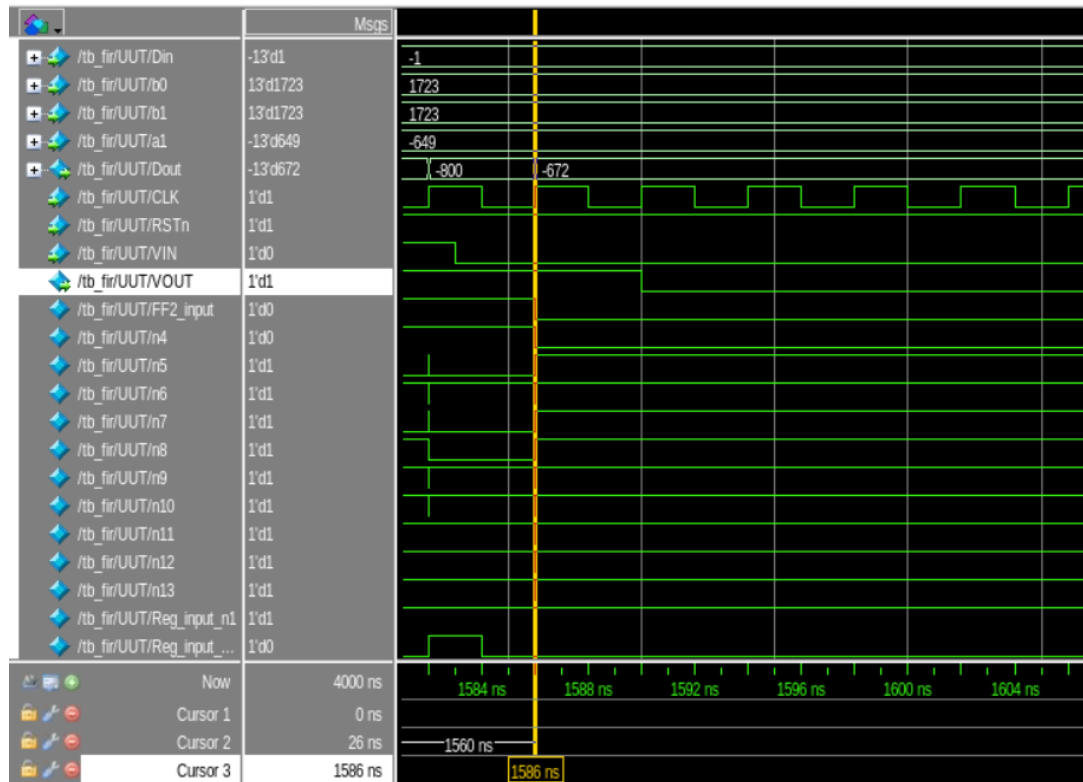


Figure 17: waveform: last high level Vout

Clock-gated architectures require smaller implementation areas because the enable signals are handled differently compared to non-clock-gated versions. In the non-clock-gated setup, an enable signal functions through a multiplexer that feeds the register output back to the input when enable is 0. In the clock-gated setup, the enable signal directly controls the clock through a simple gate. As observed in the report, the combinational area is substantial due to the inclusion of 3 multipliers and 2 adders. A more specialized architecture could further optimize the area, but would reduce flexibility. Expectedly, the clock-gated architecture consumes less power than the non_clock-gated version. Clock gating reduces dynamic power because P_{dyn} is proportional to frequency (f); clock switching is stopped when the enable signal is low, preventing the register from sampling new input values.

power estimation

Using comand as shown in Appedix4.9, we got the power estimation report in Appendix 4.13.

2.3.3 Place and Route

So start with this step, we will stop writing code whatever C, Matlab or Vhdl for a while . In one word, it's time to make "circuit schematic" into real "architecture plan" ensure that each components can find their own "place" and connects it with a "route" that will eventually lead to the actual chip. Finally, the chip layout is compact (small area), the signal transmission

efficiency is high (less time), and the energy saving is low (low power consumption). In the following we will present our data in detail.

Using the Cadence Innovus, according to the assignment file, we reduce the frequency to $f_{max}/2 = 364.963504 \text{ MHz}$ (We decided to use the min period of 5.48 ns, so the double is 10.96 ns) by modifying the .sdc file. The final structure of the filter is in Fig 31, which is the result of the innovus place-and-route process.

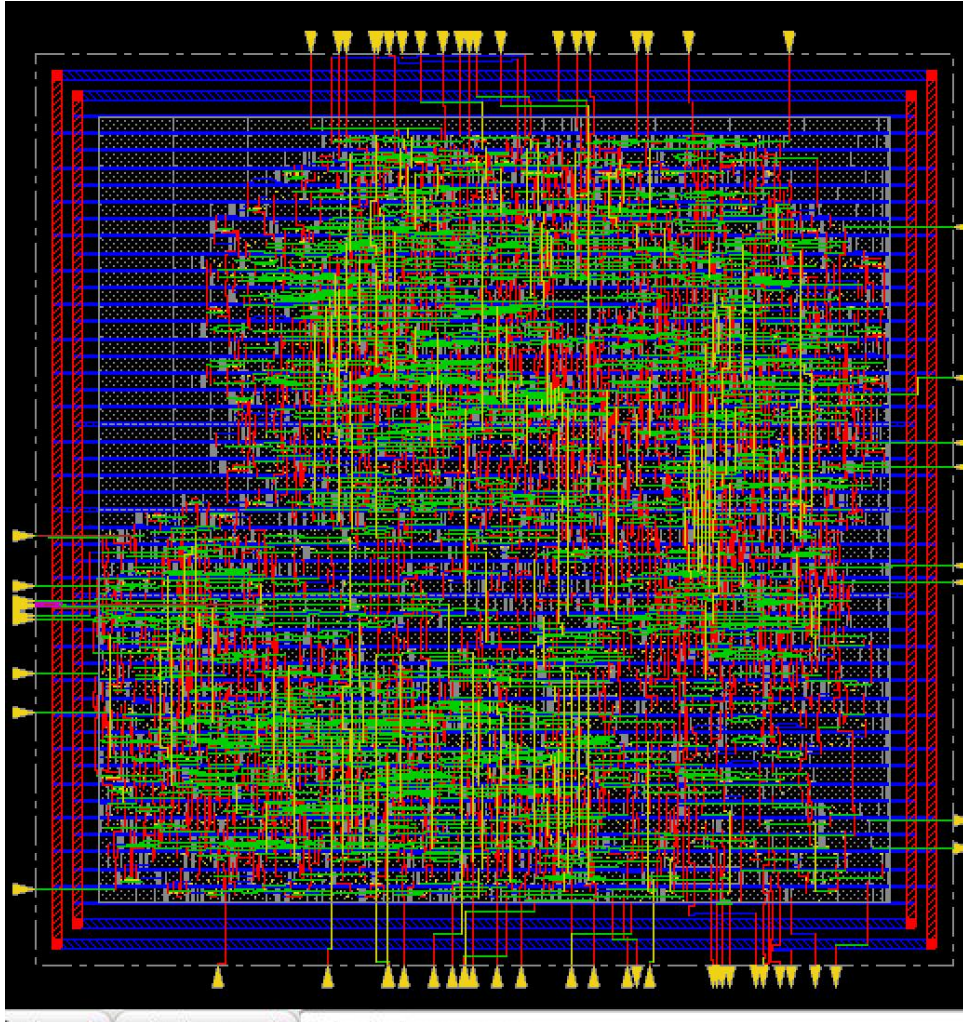


Figure 18: Layout

The terminal window tool provided the design's area metrics, with the standard cell area (Appendix 4.15) measured at $2467.7 \mu\text{m}^2$, aligning with findings from the Design Compiler. The total allocated area, which includes wiring, power rings, and the clock tree network, reaches $4034 \mu\text{m}^2$. Power consumption is reported at 0.613 mW, so 613 uW (see Appendix ??), it is double of the power estimation value found in previous synthesis (234.9579uW), because in Cadence Innovus simulation and synthesis we use the a clk period which is double of the minimum period (previous estimation is based on the minimum period). It can be seen that the power consumption is slightly less than half of what we would expect based on the linear scaling between dynamic power and frequency. The switching activity directly impacts dynamic power consumption, as it depends on the capacitance of the circuit, the supply

voltage, and the switching frequency. By analysis, the report of power can get that: Halving the clock frequency effectively reduces the switching activity and dynamic power, as evidenced by the breakdown of power in the report (dynamic power consumption is proportional to the frequency). From review in the timing_report (Appendix 4.14), the minimum period for the architecture is around the 5.48 ns.

Other End Arrival Time 0.000

- Setup 0.042

+ Phase Shift 10.960

- Uncertainty 0.070

= Required Time 10.848

- Arrival Time 4.388

= Slack Time 6.460

Then we run simulation using the fir.v (see in zip) produced by Cadence Innovus, with the period of clk equal to 10.96 ns, results are always correct.

$$y(n) = b_0x(n) + b_1x(n-1) - a_1.y(n-1) \quad (3)$$

$$y(n+1) = b_0x(n+1) + b_1x(n) - a_1.y(n) \quad (4)$$

$$y(n+1) = b_0x(n+1) + b_1x(n) - a_1[(b_0x(n) + b_1x(n-1) - a_1.y(n-1))] \quad (5)$$

$$y(n+1) = b_0x(n+1) + b_1x(n) - a_1b_0x(n) - a_1b_1x(n-1) + a_1^2.y(n-1) \quad (6)$$

$$y(n+1) = b_0x(n+1) + (b_1 - a_1b_0)x(n) - (a_1b_1)x(n-1) + a_1^2.y(n-1) \quad (7)$$

$$y(n) = b_0x(n) + (b_1 - a_1b_0)x(n-1) - (a_1b_1)x(n-2) + a_1^2.y(n-2) \quad (8)$$

$$(9)$$

This modification creates a circuit with new coefficients (coefficients a1(new), b0(new), b1(new), and b2(new)):
the coefficient of modified filter

$$a_1(\text{new}) = a_1^2 = (-0.158447265625) * (-0.158557265625) = 0.025105535984$$

$$b_0(\text{new}) = b_0 = 0.420654296875$$

$$b_1(\text{new}) = b_1 - a_1 \cdot b_0$$

$$= 0.420654296875 - (-0.158447265625 \cdot 0.420654296875)$$

$$= 0.4873046875$$

$$b_2(\text{new}) = -(a_1 * b_1) = -(-0.158447265625 * 0.420654296875) = 0.066650390625$$

for simplicity a1(new), b0(new), b1(new), b2(new) from now are called a1, b0, b1, b2. The number that would be saved in the 13-bit memory of the hardware: we must multiply the real numbers by 2^{12} , then round it to floor to get an integer number that can be stored as a binary number in 13 bits(ai and bi vectors):

$$a_{1_i} = 0.025105535984 \cdot 2^{12} = 102.83 \quad \Rightarrow \lfloor 102.83 \rfloor = 102$$

$$b_{1_i} = 0.4873046875 \cdot 2^{12} = 1996.004 \quad \Rightarrow \lfloor 1996.004 \rfloor = 1996$$

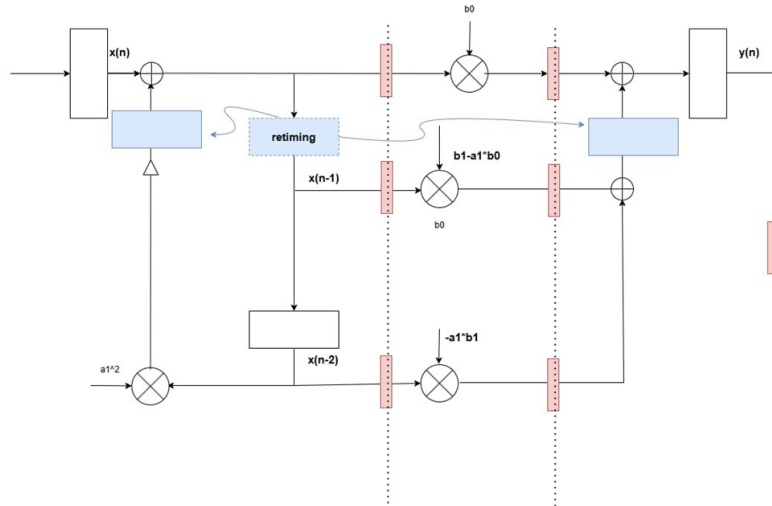
$$b_{2_i} = 0.066650390625 \cdot 2^{12} = 273.004 \quad \Rightarrow \lfloor 273.004 \rfloor = 273$$

$$b_{0_i} \text{ remains as before } 1723.$$

Now if we divide these numbers by 2 power of 12, quantized coefficients are achieved(aq and bq). Then we could see that real coefficients and quantized ones are so near to each other that means quantized error is so less. considering that in VHDL we use only ai and bi vectors .

quantized coefficient	decimal value	binary value
a1	0.02490234375	0.00000110 0110
b0	0.420654296875000	0.011010111011
b1	0.4873046875	0.011111001100
b2	0.066650390625	0.000100010001

Table 2



3.1.2 pipelining and retiming

To optimize the critical path (CP), retiming and pipelining are applied. Pipelining uses registers at feedforward cutsets (marked in pink), while retiming (marked in blue) balances delays. As shown in Figure 20, these methods reduce the CP from $2T_m + 2T_a$ to T_m , allowing higher clock frequency and improved performance.

Although the number of registers increased from 2 to 4 which means more clock cycles, the shorter CP enables better parallelism and reduces the cycle period. Additionally, to address overflow issues, the bit width was increased from 13 to 14 bits, slightly increasing hardware size but improving precision.

3.1.3 final design

In the architecture, **4 flip-flops** are used to control signal delays and synchronize registers. These flip-flops ensure proper timing and sequencing of signals in the circuit. also ensure the speed of transferring validity signal because a FF acts as a buffer.

3.2 Testing the filter and fixed point implementation

Now we must write a new C code to support the modified circuit structure. The IIR filter architecture described by the previous was then modelled into C language(Appendix 4.16).

The main task is to determine the appropriate shift amount (SHMAT) to meet the requirement of Total Harmonic Distortion (THD) below -40 dB. Testing shows:

- SHMAT=18: THD of -42.9452 dB, providing optimal performance.

Confronting the results of the new filter architecture C model (c_18_lookahead)(4.17) with the MATLAB results (mo) and the results of the C model of the previous architecture (c_17_o), we notice a slight decrease in precision. However, this decrement is acceptable. As shown in Fig22, the blue line represents the output of the new C model architecture, the red line

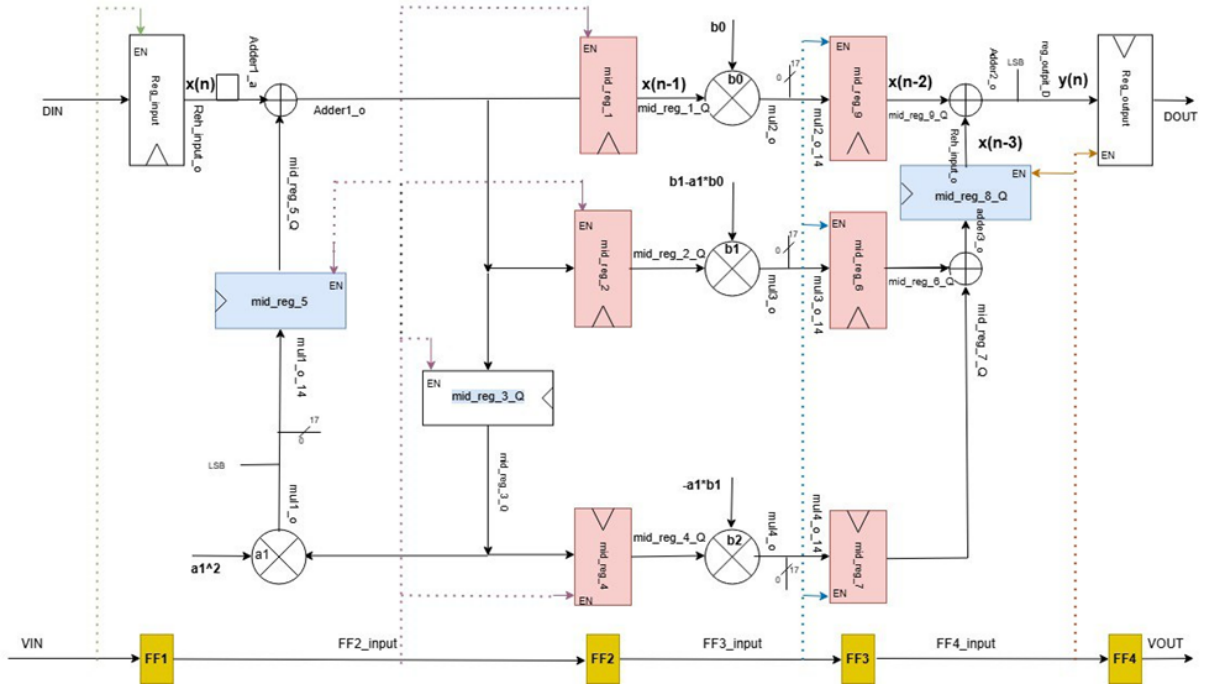


Figure 21: design of advanced architecture I.I.R filter

corresponds to the MATLAB filtering results, and the green line shows the output of the previous architecture's C model.

3.3 VLSI implementation

3.3.1 architecture explanation

Design is shown in Fig21.

In this architecture, the multiplier's output is 27-bit because one multiplicand is 13-bit and the other is 14-bit. This results in a 27-bit number with 3-bit for the integer part and 24-bit for the fractional part. Since only 14-bit are needed, a discard operation is used to discard less significant bits, leaving 2 integer bits.

Enable signal: The VIN signal is used to enable or disable the registers.

Register Disabling:

If VIN = 0: Input register sampling is disabled. -After one clock cycle, registers mid_reg_1 to mid_reg_5 are disabled. After two clock cycles, mid_reg_6 to mid_reg_9 are disabled. After three clock cycles, mid_reg_8 and output registers are disabled. After four clock cycles, the output is marked invalid, and VOUT is set to 0.

3.4 VHDL and Simulation

We build the architecture of the advanced IIR Filter then simulate it: Fig 23 is the waveform screenshot which shows that although the variation of Vin does not have influence on output results, it determines for En of input register to catch and save the input data(vin=1) and when not to take the input data(vin=0). Following this 4 clock cycles later vout get the vin signal and do the same thing.

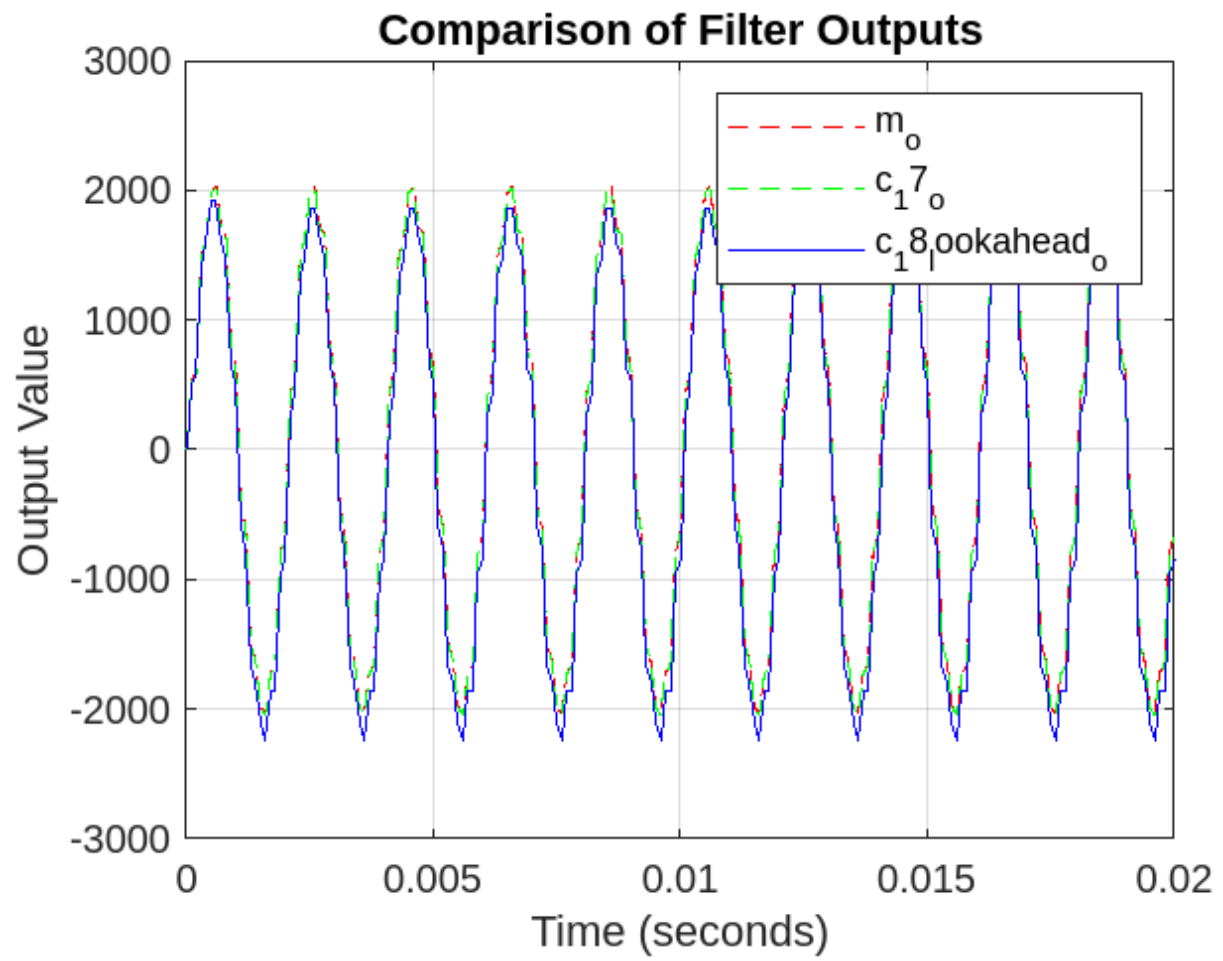


Figure 22: Comparison of $c_{18_lookahead}_o$ and m_o outputs

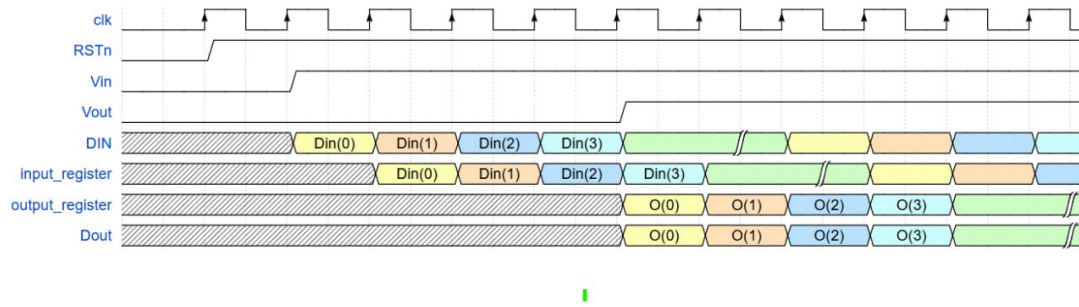


Figure 23: advanced waveform

3.4.1 Simulation Verification in Qsim

In this section, the filter architecture is simulated in Qsim using components such as `clk_generator`, `data_maker`, and `data_sink`. The testbench (`tb_fiir_lookahead`) generates output files that validate the consistency of results between the VHDL implementation and the C program. vhdl code is included in appendix 4.24.

The clock period was set to 100 000 ns (frequency = 10 kHz). it is shown in Fig 24 the amount of time required to complete the simulation (from the first rising edge of the clock sampling `VOUT='1'` to the last rising edge of the clock sampling `VOUT='1'`) is 38.806911ms. The results confirmed that the critical path delay is shorter than the clock period, allowing each circuit calculation to complete before the next clock cycle. This confirms that the clock period can be optimized further by achieving a slack of zero (`slack=0`), enabling the calculation of an ideal clock cycle for the architecture.

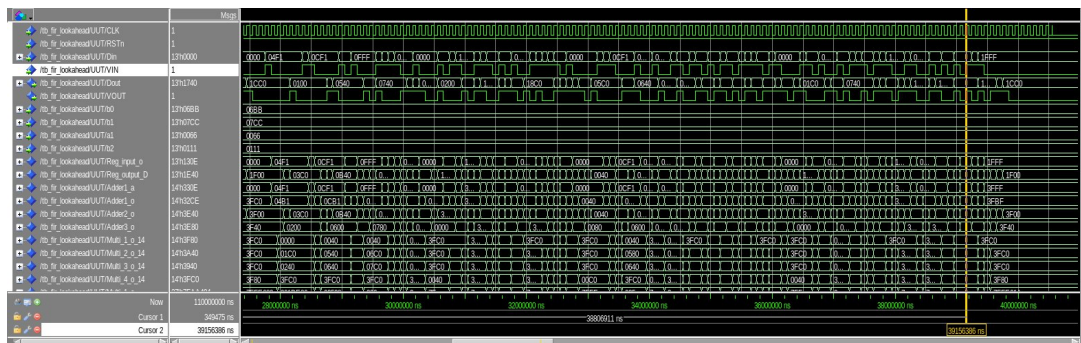


Figure 24: duration

By using schematic viewer, we obtain the top level schematic of the testbench in Fig25 bellow also The Register Transfer Level diagram of the filter is shown in Fig26

3.5 Logic Synthesis

3.5.1 Without Clock Gating

The circuit was synthesized without applying clock gating. Using the command shown in Fig4.6. The synthesis results showed:

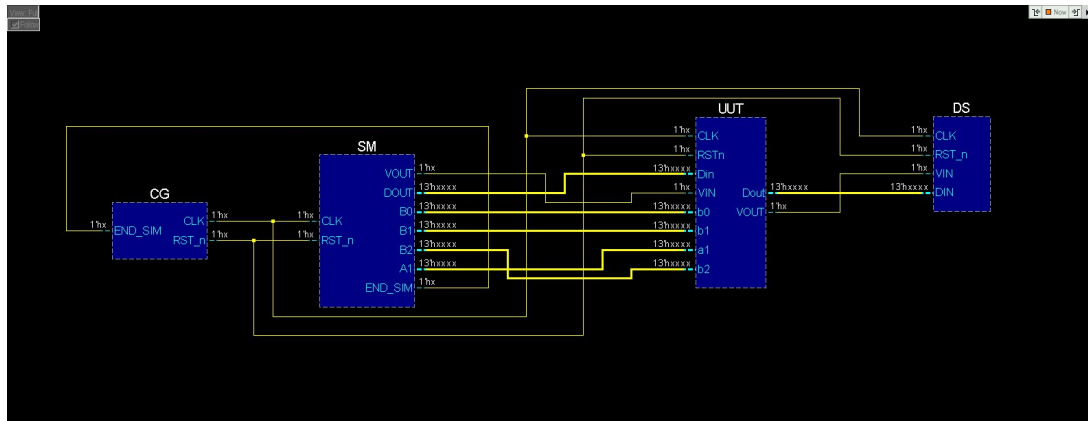


Figure 25: schematic of whole circuit (including with Testbenches)

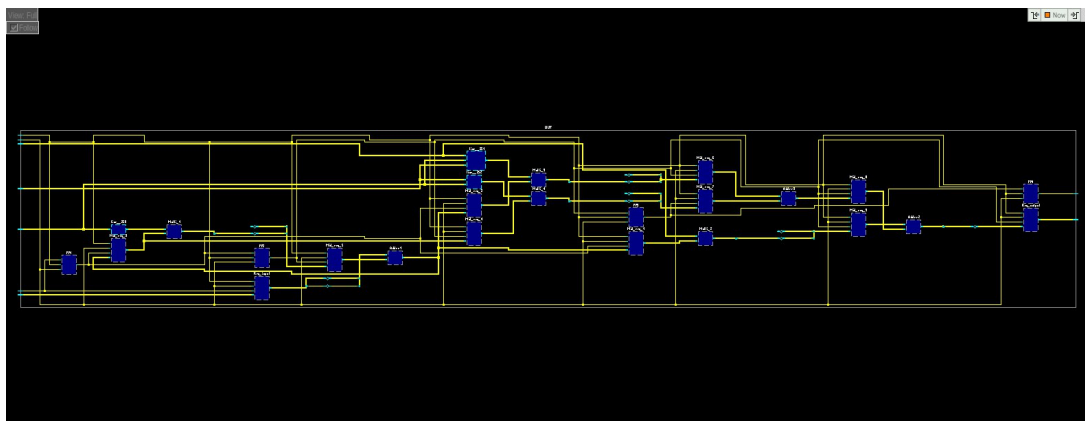


Figure 26: schematic with main testbench: UUT

- Maximum clock frequency: 555.56 MHz (corresponding to a clock delay of 1.8 ns), recorded in Appendix4.20.
- Total area: 4538.75 μm^2 , as indicated in Appendix4.18.
- Power consumption: 2.1792e+03 uW, recorded in Appendix4.19.

The synthesized design was tested by generating a new 'fir.v' file, and the outputs were validated against the C program. The time difference between the first and last valid outputs was recorded as 776 ns, consistent with simulation results. These findings are illustrated in Fig27 and Fig28.

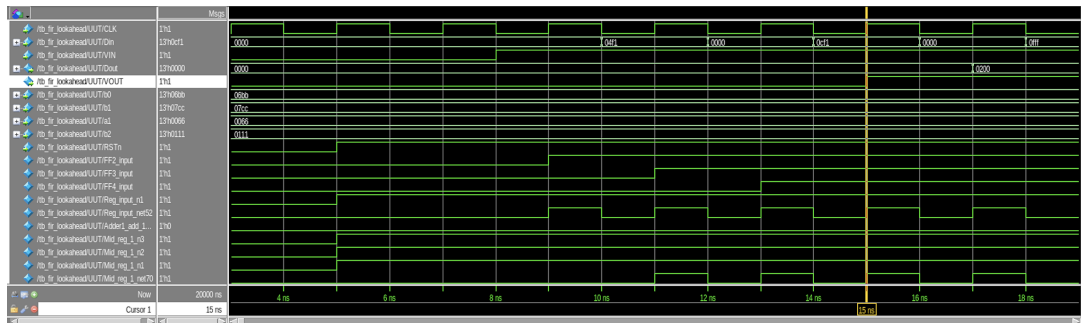


Figure 27: Waveform: first high level V_{out}

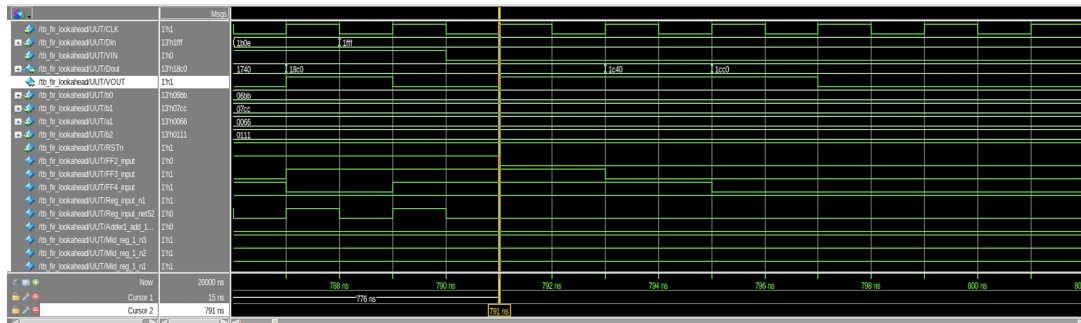


Figure 28: Waveform: last high level V_{out}

3.5.2 With Clock Gating

After enabling clock gating, the design showed significant improvements:

- Minimum clock period improved to 1.76 ns, increasing the maximum clock frequency to 568.18MHz, see in timing-report.txt.
- Total area: 4279.14 μm^2 , reduced compared to the non-clock-gating design (see in area-report.txt).
- Power consumption: 1.9652e+03 uW, lower than the design without clock gating (see in power-report.txt).

Additionally, clock gating eliminated unnecessary switching activity, improving efficiency. The new design also simplified logic by using only one multiplier instead of two multipliers and two adders, which further reduced the critical path and improved performance.

The time difference between the first and last valid outputs remained consistent with the non-clock-gating design, ensuring stability and correctness. This is demonstrated in Fig29 and Fig30.

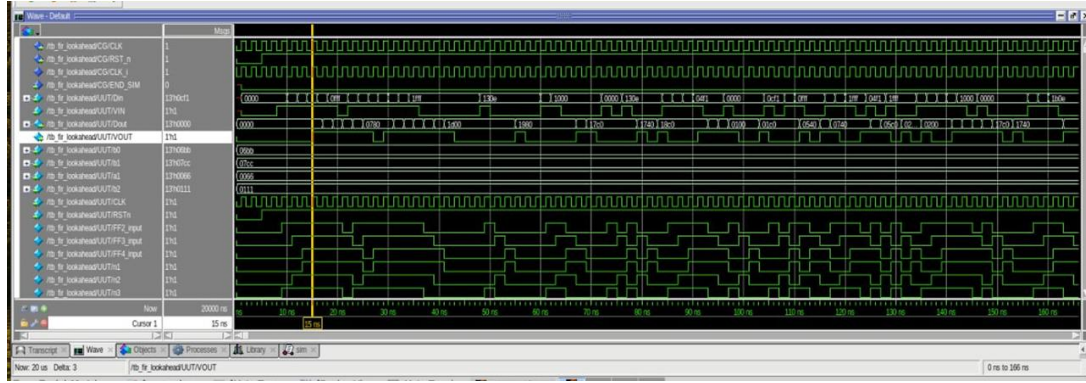


Figure 29: waveform: first high level Vout

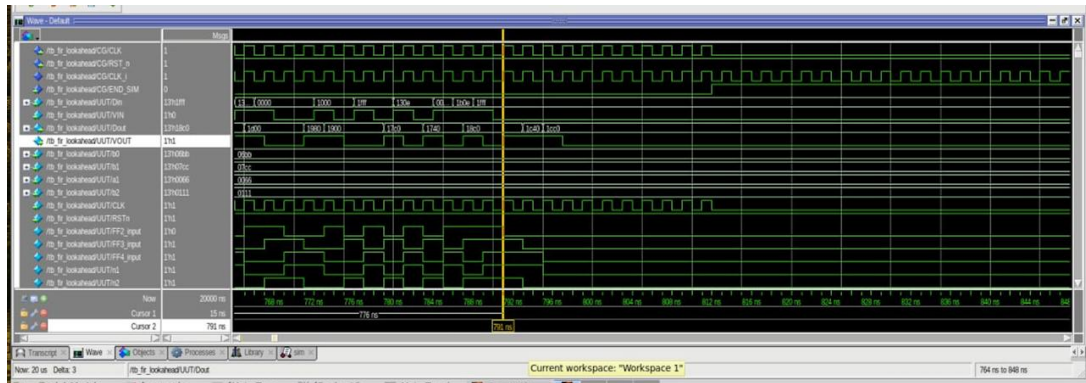


Figure 30: waveform: last high level Vout

3.5.3 Comparison of Results

Table3 summarizes the performance differences between designs with and without clock gating:

Metric	Without Clock Gating	With Clock Gating
Clock Period (ns)	1.80	1.76
Clock Frequency (MHz)	555.56	568.18
Area (μm^2)	4538.76	4279.14
Power Consumption	2.1792e+03 uW	1.9652e+03 uW
Time Difference between first Dout and last Dout (ns)	776	776

Table 3: Performance Comparison with and without Clock Gating

3.6 place and route

Using the Cadence Innovus tool, we reduced the clock frequency to $f_{\max}/2 = 284.09$ MHz, corresponding to a minimum clock period of 3.52 ns. The final layout of the filter is shown in Fig31, representing the result of the place-and-route process.

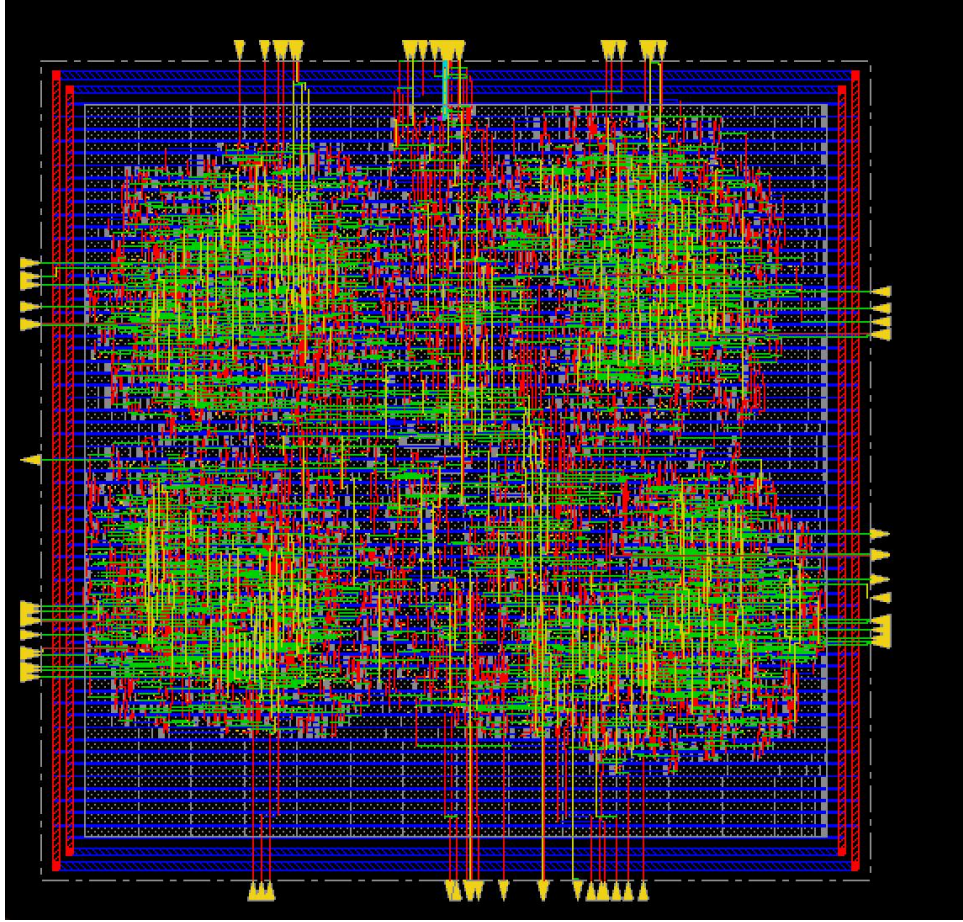


Figure 31: Final Layout after Place-and-Route Process

The design's area and power metrics are as follows:

- **Standard cell area:** $2042 \mu m^2$ (Appendix??).
- **Total allocated area (including wiring and clock tree):** $3938.864 \mu m^2$.
- **Power consumption:** 0.9518 mW (see power-innovus.txt).

By halving the clock frequency, the switching activity and dynamic power were significantly reduced, as dynamic power consumption is proportional to frequency. Timing analysis (see in ZIP file) confirms that the minimum clock period achieved by the architecture is 1.76 ns.

Path Groups: MY_{CLK} Analysis View: MyAnView Other End Arrival Time 0.036

- Setup 0.037

+ Phase Shift 3.520

- Uncertainty 0.070

= Required Time 3.449
- Arrival Time 1.385
= Slack Time 2.064

Then we run simulation using the fir-lookhead.v (it's so long so we put it into ZIP file) produced by Cadence Innovus, with the period of clk equal to 3.52 ns, results are always correct.

4 Appendix

4.1 myfilterii.c

```
1 #include<stdio.h>
2 #include<stdlib.h>
3
4 #define N 1 /// order of the filter
5 #define NTml (N) /// number of coefficients minus one (equal to the order)
6 #define NB 13 /// number of bits
7 #define SHAMT 18 /// shift amount
8
9 const int bi0 = 1723; /// coefficient b0
10 const int bi[NTml]={1723}; /// b array
11 const int ai[NTml]={-649}; /// a array
12
13 /// Perform fixed point filtering assuming direct form II
14 ///\param x is the new input sample
15 ///\return the new output sample
16 int myfilter(int x)
17 {
18     static int sw[NTml]; /// w shift register
19     static int first_run = 0; /// for cleaning the shift register
20     int i; /// index
21     int w; /// intermediate value (w)
22     int y; /// output sample
23     int fb, ff; /// feed-back and feed-forward results
24
25     /// clean the buffer
26     if (first_run == 0)
27     {
28         first_run = 1;
29         for (i=0; i<NTml; i++)
30             sw[i] = 0;
31     }
32
33     /// compute feed-back and feed-forward
34     fb = 0;
35     ff = 0;
36     for (i=0; i<NTml; i++)
37     {
38         fb -= ((sw[i]*ai[i]) >> SHAMT) << (SHAMT-NB+1);
39         ff += ((sw[i]*bi[i]) >> SHAMT) << (SHAMT-NB+1);
40     }
41
42     /// compute intermediate value (w) and output sample
43     w = x + fb;
44     y = ((w*bi0) >> SHAMT) << (SHAMT-NB+1);
45     y += ff;
46
47     /// update the shift register
48     for (i=NTml; i>0; i--)
49         sw[i] = sw[i-1];
50     sw[0] = w;
51
52     return y;
53 }
54
55 int main (int argc, char **argv)
56 {
57     FILE *fp_in;
58     FILE *fp_out;
59
60     int x;
61     int y;
62
63     /// check the command line
64     if (argc != 3)
65     {
66         printf("Use: %s <input_file> <output_file>\n", argv[0]);
67         exit(1);
68     }
69
70     /// open files
71     fp_in = fopen(argv[1], "r");
72     if (fp_in == NULL)
73     {
74         printf("Error: cannot open %s\n", argv[1]);
75         exit(2);
76     }
77     fp_out = fopen(argv[2], "w");
78
79     /// check shift amount
```

```

80  if (SHAMT < ((NB)-1))
81  {
82      printf("Error shift amount must be at least nbit-1\n");
83      exit(3);
84  }
85
86  /// get samples and apply filter
87  fscanf(fp_in, "%d", &x);
88  do{
89      y = myfilter(x);
90      fprintf(fp_out, "%d\n", y);
91      fscanf(fp_in, "%d", &x);
92  } while (!feof(fp_in));
93
94  fclose(fp_in);
95  fclose(fp_out);
96
97  return 0;
98
99 }

```

4.2 sample.txt

```
1 0
2 1265
3 0
4 3313
5 0
6 4095
7 -1
8 3313
9 -1
10 1265
11 -1
12 -1266
13 -1
14 -3314
15 -1
16 -4096
17 0
18 -3314
19 0
20 -1266
21 0
22 1265
23 0
24 3313
25 0
26 4095
27 -1
28 3313
29 -1
30 1265
31 -1
32 -1266
33 -1
34 -3314
35 -1
36 -4096
37 0
38 -3314
39 0
40 -1266
41 0
42 1265
43 0
44 3313
45 0
46 4095
47 0
48 3313
49 -1
50 1265
51 0
52 -1266
53 -1
54 -3314
55 -1
56 -4096
57 0
58 -3314
59 0
60 -1266
61 0
62 1265
63 0
64 3313
65 0
66 4095
67 0
68 3313
69 -1
70 1265
71 -1
72 -1266
73 -1
74 -3314
75 -1
76 -4096
77 0
78 -3314
79 0
80 -1266
81 0
82 1265
83 0
84 3313
```

85 0
86 4095
87 -1
88 3313
89 -1
90 1265
91 -1
92 -1266
93 0
94 -3314
95 0
96 -4096
97 0
98 -3314
99 -1
100 -1266
101 -1
102 1265
103 0
104 3313
105 -1
106 4095
107 -1
108 3313
109 -1
110 1265
111 -1
112 -1266
113 -1
114 -3314
115 -1
116 -4096
117 -1
118 -3314
119 -1
120 -1266
121 0
122 1265
123 -1
124 3313
125 0
126 4095
127 0
128 3313
129 0
130 1265
131 -1
132 -1266
133 0
134 -3314
135 0
136 -4096
137 0
138 -3314
139 -1
140 -1266
141 0
142 1265
143 0
144 3313
145 0
146 4095
147 -1
148 3313
149 -1
150 1265
151 0
152 -1266
153 -1
154 -3314
155 -1
156 -4096
157 -1
158 -3314
159 0
160 -1266
161 0
162 1265
163 -1
164 3313
165 0
166 4095
167 0
168 3313
169 0
170 1265
171 -1
172 -1266
173 -1
174 -3314
175 0
176 -4096
177 0
178 -3314
179 0

180	-1266
181	0
182	1265
183	0
184	3313
185	-1
186	4095
187	-1
188	3313
189	0
190	1265
191	0
192	-1266
193	-1
194	-3314
195	0
196	-4096
197	-1
198	-3314
199	0
200	-1266
201	-1

4.3 resultc_17.txt

```
1 0
2 512
3 576
4 1472
5 1632
6 1984
7 2016
8 1696
9 1664
10 768
11 608
12 -448
13 -608
14 -1504
15 -1632
16 -1984
17 -2048
18 -1728
19 -1664
20 -800
21 -672
22 416
23 576
24 1472
25 1632
26 1984
27 2016
28 1696
29 1664
30 768
31 608
32 -448
33 -608
34 -1504
35 -1632
36 -1984
37 -2048
38 -1728
39 -1664
40 -800
41 -672
42 416
43 576
44 1472
45 1632
46 1984
47 2016
48 1696
49 1664
50 768
51 608
52 -448
53 -608
54 -1504
55 -1632
56 -1984
57 -2048
58 -1728
59 -1664
60 -800
61 -672
62 416
63 576
64 1472
65 1632
66 1984
67 2016
68 1696
69 1664
70 768
71 608
72 -448
73 -608
74 -1504
75 -1632
76 -1984
77 -2048
78 -1728
79 -1664
80 -800
81 -672
82 416
83 576
84 1472
```

85 1632
86 1984
87 2016
88 1696
89 1664
90 768
91 608
92 -448
93 -608
94 -1504
95 -1632
96 -1984
97 -2048
98 -1728
99 -1664
100 -800
101 -672
102 416
103 576
104 1472
105 1632
106 1984
107 2016
108 1696
109 1664
110 768
111 608
112 -448
113 -608
114 -1504
115 -1632
116 -1984
117 -2048
118 -1728
119 -1664
120 -800
121 -672
122 416
123 576
124 1472
125 1632
126 1984
127 2016
128 1696
129 1664
130 768
131 608
132 -448
133 -608
134 -1504
135 -1632
136 -1984
137 -2048
138 -1728
139 -1664
140 -800
141 -672
142 416
143 576
144 1472
145 1632
146 1984
147 2016
148 1696
149 1664
150 768
151 608
152 -448
153 -608
154 -1504
155 -1632
156 -1984
157 -2048
158 -1728
159 -1664
160 -800
161 -672
162 416
163 576
164 1472
165 1632
166 1984
167 2016
168 1696
169 1664
170 768
171 608
172 -448
173 -608
174 -1504
175 -1632
176 -1984
177 -2048
178 -1728
179 -1664

180	-800
181	-672
182	416
183	576
184	1472
185	1632
186	1984
187	2016
188	1696
189	1664
190	768
191	608
192	-448
193	-608
194	-1504
195	-1632
196	-1984
197	-2048
198	-1728
199	-1664
200	-800
201	-672

4.4 thd filter

```
1 % Read the input files
2 resultm_data = load('resultsm.txt'); % Load simple.txt file
3 resultc_data = load('resultc_17.txt'); % Load resultc_I.txt file
4 fs=10000;
5 n=length(resultc_data);
6 fft_resultc=fft(resultc_data);
7 f = (0:n-1)*(fs/n); % Frequency vector
8
9 % Step 4: Magnitude of FFT
10 magnitude = abs(fft_resultc/n); % Normalize and take magnitude
11
12 % Step 5: Plot the FFT (only first half due to symmetry)
13 figure;
14 plot(f(1:n/2), magnitude(1:n/2)); % Plot only the positive half
15 title('FFT of the Data');
16 xlabel('Frequency (Hz)');
17 ylabel('Magnitude');
18 % Ensure that the files are properly loaded as vectors
19 if ~isvector(resultm_data) || ~isvector(resultc_data)
20     error('Data files must contain vectors.');
```

```
21 end
22
23
24
25 % Define the sampling frequency (fs), which is needed for THD calculation.
26 % You must know the sampling frequency (in Hz) of the signals.
27 % For example, assume the sampling frequency is 1000 Hz:
28 %fs = 10000; % Change this value to the correct sampling frequency
29
30 % Calculate THD for both signals using MATLAB's thd function
31 % The thd function expects the input signal and sampling frequency
32
33 % THD of the original signal (simple.txt)
34 %thd_simple = thd(simple_data, fs);
35
36 % THD of the filtered signal (resultc_I.txt)
37 %thd_resultc = thd(resultc_data, fs);
38
39 % Display the THD values
40 %fprintf('THD of simple.txt: %.2f%%\n', thd_simple);
41 %fprintf('THD of resultc_I.txt: %.2f%%\n', thd_resultc);
42
43 % Optionally, plot both signals for comparison
44 figure;
45 subplot(2,1,1);
46 plot(resultm_data);
47 title('resultsm');
48 xlabel('Sample Number');
49 ylabel('Amplitude');
50
51 subplot(2,1,2);
52 plot(resultc_data);
53 title('Filtered Signal (resultc_I.txt)');
54 xlabel('Sample Number');
55 ylabel('Amplitude');
56
57
58 figure;
59 plot(resultc_data,'r');
60 hold on
61 plot(resultm_data,'b:');
62 %% calculation thd
63 THD_filter_II_c=thd(resultc_data,10000);
```

4.5 fir.vhd

```
1 library IEEE;
2 use IEEE.std_logic_1164.all;
3 use IEEE.numeric_std.all;
4
5 entity fiir is
6   generic (
7     N: integer :=13;
8     SHAMT: integer :=17);
9   port (
10    CLK : in std_logic ;
11    RSTn : in std_logic ;
12    Din : in std_logic_vector((N-1) downto 0);
13    VIN : in std_logic ;
14    b0 : in std_logic_vector((N-1) downto 0);
15    b1 : in std_logic_vector((N-1) downto 0);
16    a0 : in std_logic_vector((N-1) downto 0);
17    Dout :out std_logic_vector((N-1) downto 0);
18    VOUT :out std_logic);
19 end fiir;
20
21
22 architecture behavioral of fiir is
23
24   component multiplier
25     generic(N:integer);
26     port(
27       a: in std_logic_vector(N-1 downto 0);
28       b: in std_logic_vector(N downto 0);
29       o: out std_logic_vector(2*N downto 0));
30   end component;
31
32   component adder
33     generic(N:integer);
34     port(
35       a: in std_logic_vector(N downto 0);
36       b: in std_logic_vector(N downto 0);
37       o: out std_logic_vector(N downto 0));
38   end component;
39
40   component DFF_syn
41     port (Din,CK,RSTn : in std_logic ;
42           Q :out std_logic );
43   end component;
44
45   component generic_reg
46     generic(N:integer);
47     Port ( CLK : in STD_LOGIC;
48           rst_n : in STD_LOGIC;
49           en : in STD_LOGIC;
50           D : in STD_LOGIC_VECTOR(N-1 downto 0);
51           Q : out STD_LOGIC_VECTOR(N-1 downto 0));
52   end component;
53
54   --signal
55
56   signal Reg_input_o, Reg_output_D:std_logic_vector((N-1) downto 0);
57   signal Adder1_a, Adder1_b, Adder1_o, Adder2_a, Adder2_b, Adder2_o: std_logic_vector(N downto 0);
58   signal Multi_2_input, Multi_3_input: std_logic_vector(N downto 0);
59   signal Multi_1_o, Multi_2_o, Multi_3_o: std_logic_vector(2*N downto 0);
60   signal FF2_input: std_logic;
61   signal Mid_reg_D, mid_reg_Q: std_logic_vector(N downto 0);
62   begin
63
64     Reg_input: generic_reg
65       generic map(N=>N)
66       port map(
67         CLK=>CLK,
68         rst_n=>RSTn,
69         en=>VIN,
70         D=>Din,
71         Q=>Reg_input_o);
72
73     Reg_output: generic_reg
74       generic map(N=>N)
75       port map(
76         CLK=>CLK,
77         rst_n=>RSTn,
78         en=>FF2_input,
79         D=>Reg_output_D,
80         Q=>Dout);
81
82     Adder1: adder
83       generic map(N=>N)
84       port map(
```

```

85     a=>Adder1_a,
86     b=>Adder1_b,
87     o=>Adder1_o);
88
89 Adder2: adder
90     generic map(N=>N)
91     port map(
92         a=>Adder2_a,
93         b=>Adder2_b,
94         o=>Adder2_o);
95
96 Mid_reg: generic_reg
97     generic map(N=>(N+1))
98     port map(
99         CLK=>CLK,
100        rst_n=>RSTn,
101        en=> FF2_input,
102        D=>Mid_reg_D,
103        Q=>Mid_reg_Q);
104
105 Multi_1: multiplier
106     generic map(N=>N)
107     port map(
108         a=>a0, --13
109         b=>Mid_reg_Q, --14
110         o=>Multi_1_o); --27
111
112 Multi_2: multiplier
113     generic map(N=>N)
114     port map(
115         a=>b0, --13
116         b=>Multi_2_input, --14
117         o=>Multi_2_o); --27
118
119 Multi_3: multiplier
120     generic map(N=>N)
121     port map(
122         a=>b1, --13
123         b=>Multi_3_input, --14
124         o=>Multi_3_o); --27
125
126 FF1: DFF_syn
127     port map(Din=> VIN,
128         CK=>CLK,
129         RSTn=>RSTn,
130         Q=>FF2_input);
131
132 FF2: DFF_syn
133     port map(Din=> FF2_input,
134         CK=>CLK,
135         RSTn=>RSTn,
136         Q=>VOUT);
137
138
139 Adder1_a<= Reg_input_o(N-1)&Reg_input_o;
140 Mid_reg_D<= Adder1_o;
141 Multi_2_input<= Adder1_o;
142 Adder1_b<= Multi_1_o((2*N-1) downto SHAMT)& "00000"; --number of zeroes =(SHAMT-N)+1=17-13+1=5
143 Multi_3_input<= Mid_reg_Q;
144 Adder2_b<= Multi_3_o((2*N-1) downto SHAMT)& "00000";
145 Adder2_a<= Multi_2_o((2*N-1) downto SHAMT)& "00000";
146 Reg_output_D<= Adder2_o((N-1) downto 0);
147
148
149 end behavioral;

```

4.6 ad_comand_synthesis.txt

```
1 source /eda/scripts/init_design_vision
2
3 design_vision
4 #dc_shell-xg-t
5 # read source files;
6 analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/adder.vhd
7 analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/DFF_syn.vhd
8 analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/fiir_lookahead.vhd
9 analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/generic_reg.vhd
10 analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/multiplier.vhd
11 #analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/clk_gen.vhd
12 #analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/data_maker.vhd
13 #analyze -f vhd1 -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/design/data_sink.vhd
14 #analyze -f verilog -lib WORK /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/src/tb/tb_fir_lookahead.v
15
16 set power_preserve_rtl_hier_names true
17
18 elaborate fiir_lookahead -arch behavioral -lib WORK
19
20 uniquify
21
22 link
23 # applying constraint
24
25 #create_clock -name MY_CLK -period 0 CLK
26 create_clock -name MY_CLK -period 1.76 CLK
27 #create_clock -name MY_CLK -period 1.26 CLK
28
29 set_dont_touch_network MY_CLK
30
31 set_clock_uncertainty 0.07 [get_clocks MY_CLK]
32 #set_clock_uncertainty 0.007 [get_clocks MY_CLK]
33
34 set_input_delay 0.5 -max -clock MY_CLK [remove_from_collection [all_inputs] CLK]
35 #set_input_delay 0.05 -max -clock MY_CLK [remove_from_collection [all_inputs] CLK]
36
37 set_output_delay 0.5 -max -clock MY_CLK [all_outputs]
38 #set_output_delay 0.05 -max -clock MY_CLK [all_outputs]
39
40 set OLOAD [load_of NangateOpenCellLibrary/BUF_X4/A]
41 set_load $OLOAD [all_outputs]
42
43 #compile
44 compile -gate_clock
45
46 report_timing >> /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/syn/timing_CLK_GATE.txt
47 report_area >> /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/syn/area_CLK_GATE.txt
48
49
50 # save the information for power estimation
51
52 ungroup -all -flatten
53
54 change_names -hierarchy -rules verilog
55
56 write_sdf /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/syn/netlist/fiir_lookahead.sdf
57
58 write -f verilog -hierarchy -output /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/syn/netlist/fiir_lookahead.v
59
60 write_sdc /home/isa05_2024_2025/Desktop/milad/lab01_Advanced/syn/netlist/fiir_lookahead.sdc
61 quit
```

4.7 area_without_clk_gate.txt

```
1
2 *****
3 Report : area
4 Design : fiir
5 Version: S-2021.06-SP4
6 Date   : Sat Nov 16 16:12:50 2024
7 *****
8
9 Library(s) Used:
10
11   NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
12
13 Number of ports:          662
14 Number of nets:          2272
15 Number of cells:         1355
16 Number of combinational cells: 1290
17 Number of sequential cells:    42
18 Number of macros/black boxes:    0
19 Number of buf/inv:          147
20 Number of references:       17
21
22 Combinational area:      2367.932007
23 Buf/Inv area:           84.322000
24 Noncombinational area:   223.440007
25 Macro/Black Box area:    0.000000
26 Net Interconnect area:   undefined (Wire load has zero net area)
27
28 Total cell area:         2591.372015
29 Total area:             undefined
30 1
```


4.8 timing_without_clk_gate.txt

```

1 Information: Updating design information... (UID-85)
2
3 *****
4 Report : timing
5     -path full
6     -delay max
7     -max_paths 1
8 Design : fiir
9 Version: S-2021.06-SP4
10 Date   : Sat Nov 16 16:12:50 2024
11 *****
12
13 Operating Conditions: typical      Library: NangateOpenCellLibrary
14 Wire Load Model Mode: top
15
16 Startpoint: al[1] (input port clocked by MY_CLK)
17 Endpoint: Reg_output/Q_reg[12]
18           (rising edge-triggered flip-flop clocked by MY_CLK)
19 Path Group: MY_CLK
20 Path Type: max
21
22 Des/Clust/Port      Wire Load Model      Library
23 -----
24 fiir                5K_hvratio_1_1      NangateOpenCellLibrary
25
26 Point                                     Incr      Path
27 -----
28 clock MY_CLK (rise edge)                  0.00      0.00
29 clock network delay (ideal)                0.00      0.00
30 input external delay                      0.50      0.50 r
31 al[1] (in)                                0.00      0.50 r
32 Multi_1/a[1] (multiplier_N13_2)            0.00      0.50 r
33 Multi_1/mult_16/a[1] (multiplier_N13_2_DW_mult_tc_0_DW_mult_tc_1)
34                                         0.00      0.50 r
35 Multi_1/mult_16/U438/Z (XOR2_X1)           0.11      0.61 r
36 Multi_1/mult_16/U425/ZN (OR2_X2)           0.08      0.70 r
37 Multi_1/mult_16/U510/ZN (AOI222_X1)        0.06      0.75 f
38 Multi_1/mult_16/U108/S (HA_X1)            0.08      0.83 f
39 Multi_1/mult_16/U692/ZN (AOI222_X1)        0.11      0.94 r
40 Multi_1/mult_16/U463/ZN (INV_X1)           0.03      0.97 f
41 Multi_1/mult_16/U691/ZN (AOI222_X1)        0.09      1.06 r
42 Multi_1/mult_16/U468/ZN (INV_X1)           0.03      1.09 f
43 Multi_1/mult_16/U690/ZN (AOI222_X1)        0.09      1.19 r
44 Multi_1/mult_16/U459/ZN (INV_X1)           0.03      1.21 f
45 Multi_1/mult_16/U689/ZN (AOI222_X1)        0.09      1.31 r
46 Multi_1/mult_16/U466/ZN (INV_X1)           0.03      1.33 f
47 Multi_1/mult_16/U688/ZN (AOI222_X1)        0.09      1.43 r
48 Multi_1/mult_16/U467/ZN (INV_X1)           0.03      1.45 f
49 Multi_1/mult_16/U687/ZN (AOI222_X1)        0.09      1.55 r
50 Multi_1/mult_16/U456/ZN (INV_X1)           0.03      1.58 f
51 Multi_1/mult_16/U686/ZN (AOI222_X1)        0.09      1.67 r
52 Multi_1/mult_16/U460/ZN (INV_X1)           0.03      1.70 f
53 Multi_1/mult_16/U685/ZN (AOI222_X1)        0.09      1.79 r
54 Multi_1/mult_16/U462/ZN (INV_X1)           0.03      1.82 f
55 Multi_1/mult_16/U684/ZN (AOI222_X1)        0.09      1.91 r
56 Multi_1/mult_16/U464/ZN (INV_X1)           0.03      1.94 f
57 Multi_1/mult_16/U683/ZN (AOI222_X1)        0.09      2.03 r
58 Multi_1/mult_16/U465/ZN (INV_X1)           0.03      2.06 f
59 Multi_1/mult_16/U682/ZN (AOI222_X1)        0.11      2.16 r
60 Multi_1/mult_16/U681/ZN (AOI222_X1)        0.07      2.23 f
61 Multi_1/mult_16/U680/ZN (AOI222_X1)        0.11      2.34 r
62 Multi_1/mult_16/U679/ZN (AOI222_X1)        0.07      2.41 f
63 Multi_1/mult_16/U13/CO (FA_X1)             0.10      2.51 f
64 Multi_1/mult_16/U12/S (FA_X1)             0.13      2.64 r
65 Multi_1/mult_16/product[18] (multiplier_N13_2_DW_mult_tc_0_DW_mult_tc_1)
66                                         0.00      2.64 r
67 Multi_1/o[18] (multiplier_N13_2)           0.00      2.64 r
68 U16/ZN (INV_X1)                           0.02      2.66 f
69 U28/Z (XOR2_X1)                           0.07      2.73 f
70 Adder1/b[6] (adder_N13_1)                 0.00      2.73 f
71 Adder1/add_16/B[6] (adder_N13_1_DW01_add_0_DW01_add_1)
72                                         0.00      2.73 f
73 Adder1/add_16/U1_6/S (FA_X1)              0.19      2.92 r
74 Adder1/add_16/SUM[6] (adder_N13_1_DW01_add_0_DW01_add_1)
75                                         0.00      2.92 r
76 Adder1/o[6] (adder_N13_1)                 0.00      2.92 r
77 Multi_2/b[6] (multiplier_N13_1)           0.00      2.92 r
78 Multi_2/mult_16/b[6] (multiplier_N13_1_DW_mult_tc_0_DW_mult_tc_2)
79                                         0.00      2.92 r
80 Multi_2/mult_16/U502/ZN (XNOR2_X1)         0.06      2.98 f
81 Multi_2/mult_16/U499/ZN (AOI222_X1)        0.08      3.06 r
82 Multi_2/mult_16/U446/ZN (XNOR2_X1)        0.07      3.13 r
83 Multi_2/mult_16/U103/S (FA_X1)            0.12      3.25 f
84 Multi_2/mult_16/U695/ZN (AOI222_X1)        0.11      3.36 r

```

85	Multi_2/mult_16/U473/ZN (INV_X1)	0.03	3.39	f
86	Multi_2/mult_16/U694/ZN (AOI222_X1)	0.09	3.48	r
87	Multi_2/mult_16/U469/ZN (INV_X1)	0.03	3.51	f
88	Multi_2/mult_16/U693/ZN (AOI222_X1)	0.09	3.60	r
89	Multi_2/mult_16/U472/ZN (INV_X1)	0.03	3.63	f
90	Multi_2/mult_16/U692/ZN (AOI222_X1)	0.09	3.72	r
91	Multi_2/mult_16/U470/ZN (INV_X1)	0.03	3.75	f
92	Multi_2/mult_16/U691/ZN (AOI222_X1)	0.09	3.84	r
93	Multi_2/mult_16/U445/ZN (INV_X1)	0.03	3.87	f
94	Multi_2/mult_16/U690/ZN (AOI222_X1)	0.09	3.96	r
95	Multi_2/mult_16/U471/ZN (INV_X1)	0.03	3.99	f
96	Multi_2/mult_16/U689/ZN (AOI222_X1)	0.09	4.08	r
97	Multi_2/mult_16/U474/ZN (INV_X1)	0.03	4.11	f
98	Multi_2/mult_16/U688/ZN (AOI222_X1)	0.11	4.22	r
99	Multi_2/mult_16/U687/ZN (AOI222_X1)	0.07	4.28	f
100	Multi_2/mult_16/U686/ZN (AOI222_X1)	0.11	4.40	r
101	Multi_2/mult_16/U685/ZN (AOI222_X1)	0.07	4.46	f
102	Multi_2/mult_16/U13/CO (FA_X1)	0.10	4.56	f
103	Multi_2/mult_16/U12/CO (FA_X1)	0.09	4.65	f
104	Multi_2/mult_16/U11/CO (FA_X1)	0.09	4.74	f
105	Multi_2/mult_16/U10/CO (FA_X1)	0.09	4.83	f
106	Multi_2/mult_16/U9/S (FA_X1)	0.11	4.94	f
107	Multi_2/mult_16/product[21] (multiplier_N13_1_DW_mult_tc_0_DW_mult_tc_2)			
108		0.00	4.94	f
109	Multi_2/o[21] (multiplier_N13_1)	0.00	4.94	f
110	Adder2/a[9] (adder_N13_0)	0.00	4.94	f
111	Adder2/add_16/A[9] (adder_N13_0_DW01_add_0)	0.00	4.94	f
112	Adder2/add_16/U1_9/CO (FA_X1)	0.10	5.04	f
113	Adder2/add_16/U1_10/CO (FA_X1)	0.09	5.13	f
114	Adder2/add_16/U1_11/CO (FA_X1)	0.09	5.23	f
115	Adder2/add_16/U1_12/S (FA_X1)	0.13	5.36	r
116	Adder2/add_16/SUM[12] (adder_N13_0_DW01_add_0)	0.00	5.36	r
117	Adder2/o[12] (adder_N13_0)	0.00	5.36	r
118	Reg_output/D[12] (generic_reg_N13_0)	0.00	5.36	r
119	Reg_output/U33/ZN (NAND2_X1)	0.03	5.38	f
120	Reg_output/U32/ZN (OAI21_X1)	0.03	5.41	r
121	Reg_output/Q_reg[12]/D (DFFR_X1)	0.01	5.42	r
122	data arrival time		5.42	
123				
124	clock MY_CLK (rise edge)	5.53	5.53	
125	clock network delay (ideal)	0.00	5.53	
126	clock uncertainty	-0.07	5.46	
127	Reg_output/Q_reg[12]/CK (DFFR_X1)	0.00	5.46	r
128	library setup time	-0.04	5.42	
129	data required time		5.42	
130	-----			
131	data required time		5.42	
132	data arrival time		-5.42	
133	-----			
134	slack (MET)		0.00	
135				
136				
137	1			

4.9 comand_power_estimation.txt

```
1 # launch Questasim with proper option
2 source /eda/scripts/init_questa_core_prime
3 vlib work
4 #vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/adder.vhd
5 vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/clk_gen.vhd
6 vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/data_maker.vhd
7 vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/data_sink.vhd
8 #vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/DFE_syn.vhd
9 #vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/fir.vhd
10 #vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/generic_reg.vhd
11 #vcom -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/multiplier.vhd
12 vlog -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/src/tb/tb_fir.v
13 vlog -work ./work /home/isa05_2024_2025/Desktop/monica/lab01/syn/netlist/fir.v
14 #vsim -L /eda/dk/nangate45/verilog/qsim2020.4 work.tb_fir
15 vsim -L /eda/dk/nangate45/verilog/qsim2020.4 -sdftyp /tb_fir/UUT=/home/isa05_2024_2025/Desktop/monica/lab01/syn/netlist/fir.sdf work.tb_fir
16
17 vcd file /home/isa05_2024_2025/Desktop/monica/lab01/simPower/work/vcd/fir_syn.vcd
18 vcd add /tb_fir/UUT/*
19 add wave -position insertpoint sim:/tb_fir/UUT/*
20 run 2 us
21
22 ## convert the file with the switching activity launch syn. design compile
23 vcd2saif -input /home/isa05_2024_2025/Desktop/monica/lab01/
24 simPower/work/vcd/fir_syn.vcd -output /home/isa05_2024_2025/Desktop/monica/lab01/simPower/work/saif/myfir_syn.saif
25
26 ##power consumption estimation
27 read_verilog -netlist /home/isa05_2024_2025/Desktop/monica/lab01/syn/netlist/fir.v
28 current_design fir
29 read_saif -input /home/isa05_2024_2025/Desktop/monica/lab01/simPower/work/saif/myfir_syn.saif -instance tb_fir/UUT -unit ns -scale 1.
30
31 create_clock -name MY_CLK -period 5.53 CLK
32
33 report_power >> /home/isa05_2024_2025/Desktop/monica/lab01/syn/power_withoutCLK_GATE.txt
```

4.10 power_withoutCLK_GATE.txt

```

1 Information: Updating design information... (UID-85)
2 Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
3 Warning: The derived toggle rate value (0.361664) for the clock net 'CLK' conflicts with the annotated value (0.333607). Using the annotated value
4
5 *****
6 Report : power
7         -analysis_effort low
8 Design : fiir
9 Version: S-2021.06-SP4
10 Date   : Sat Nov 16 17:12:33 2024
11 *****
12
13
14 Library(s) Used:
15
16     NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
17
18
19 Operating Conditions: typical    Library: NangateOpenCellLibrary
20 Wire Load Model Mode: top
21
22 Design      Wire Load Model      Library
23 -----
24 fiir        5K_hvratio_1_1      NangateOpenCellLibrary
25
26
27 Global Operating Voltage = 1.1
28 Power-specific unit information :
29     Voltage Units = 1V
30     Capacitance Units = 1.000000ff
31     Time Units = 1ns
32     Dynamic Power Units = 1uW      (derived from V,C,T units)
33     Leakage Power Units = 1nW
34
35
36 Cell Internal Power   = 199.2074 uW   (59%)
37 Net Switching Power  = 137.7065 uW   (41%)
38 -----
39 Total Dynamic Power   = 336.9139 uW   (100%)
40
41 Cell Leakage Power    = 52.1824 uW
42
43
44
45 Power Group      Internal      Switching      Leakage      Total
46 -----          Power        Power          Power        Power ( % ) Attrs
47 io_pad           0.0000         0.0000         0.0000         0.0000 ( 0.00%)
48 memory           0.0000         0.0000         0.0000         0.0000 ( 0.00%)
49 black_box        0.0000         0.0000         0.0000         0.0000 ( 0.00%)
50 clock_network    0.0000         0.0000         0.0000         0.0000 ( 0.00%)
51 register         43.9092         16.4767        3.6953e+03      64.0812 ( 16.47%)
52 sequential       0.0000         0.0000         0.0000         0.0000 ( 0.00%)
53 combinational    155.2982        121.2298       4.8487e+04      325.0151 ( 83.53%)
54 -----
55 Total            199.2074 uW    137.7065 uW    5.2182e+04 nW   389.0963 uW
56 1

```

4.11 area_clk_gate.txt

```
1
2 *****
3 Report : area
4 Design : fiir
5 Version: S-2021.06-SP4
6 Date   : Sat Nov 16 20:03:27 2024
7 *****
8
9 Library(s) Used:
10
11   NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
12
13 Number of ports:          674
14 Number of nets:          2126
15 Number of cells:         1233
16 Number of combinational cells: 1159
17 Number of sequential cells:   45
18 Number of macros/black boxes:  0
19 Number of buf/inv:         133
20 Number of references:       17
21
22 Combinational area:      2262.596009
23 Buf/Inv area:           73.150000
24 Noncombinational area:   235.410007
25 Macro/Black Box area:    0.000000
26 Net Interconnect area:   undefined (Wire load has zero net area)
27
28 Total cell area:         2498.006017
29 Total area:              undefined
30 1
```

4.12 timing_clk_gate.txt

```

1 Information: Updating design information... (UID=85)
2
3 *****
4 Report : timing
5     -path full
6     -delay max
7     -max_paths 1
8 Design : fiir
9 Version: S-2021.06-SP4
10 Date   : Sat Nov 16 20:03:27 2024
11 *****
12
13 Operating Conditions: typical      Library: NangateOpenCellLibrary
14 Wire Load Model Mode: top
15
16 Startpoint: al[1] (input port clocked by MY_CLK)
17 Endpoint: Reg_output/Q_reg[12]
18           (rising edge-triggered flip-flop clocked by MY_CLK)
19 Path Group: MY_CLK
20 Path Type: max
21
22 Des/Clust/Port      Wire Load Model      Library
23 -----
24 fiir                5K_hvrat1o_1_1      NangateOpenCellLibrary
25
26 Point                                     Incr      Path
27 -----
28 clock MY_CLK (rise edge)                 0.00      0.00
29 clock network delay (ideal)               0.00      0.00
30 input external delay                     0.50      0.50 r
31 al[1] (in)                               0.00      0.50 r
32 Multi_1/a[1] (multiplier_N13_2)          0.00      0.50 r
33 Multi_1/mult_16/a[1] (multiplier_N13_2_DW_mult_tc_0_DW_mult_tc_1)
34                                     0.00      0.50 r
35 Multi_1/mult_16/U682/Z (XOR2_X1)          0.11      0.61 r
36 Multi_1/mult_16/U437/ZN (OR2_X2)          0.08      0.70 r
37 Multi_1/mult_16/U493/ZN (AOI222_X1)       0.06      0.75 f
38 Multi_1/mult_16/U108/S (HA_X1)           0.08      0.83 f
39 Multi_1/mult_16/U439/ZN (INV_X1)          0.03      0.86 r
40 Multi_1/mult_16/U441/ZN (AOI222_X1)       0.06      0.92 f
41 Multi_1/mult_16/U674/ZN (AOI222_X1)       0.10      1.02 r
42 Multi_1/mult_16/U450/ZN (INV_X1)          0.03      1.04 f
43 Multi_1/mult_16/U673/ZN (AOI222_X1)       0.09      1.14 r
44 Multi_1/mult_16/U448/ZN (INV_X1)          0.03      1.16 f
45 Multi_1/mult_16/U672/ZN (AOI222_X1)       0.09      1.26 r
46 Multi_1/mult_16/U449/ZN (INV_X1)          0.03      1.29 f
47 Multi_1/mult_16/U671/ZN (AOI222_X1)       0.09      1.38 r
48 Multi_1/mult_16/U434/ZN (INV_X1)          0.03      1.41 f
49 Multi_1/mult_16/U670/ZN (AOI222_X1)       0.09      1.50 r
50 Multi_1/mult_16/U443/ZN (INV_X1)          0.03      1.53 f
51 Multi_1/mult_16/U669/ZN (AOI222_X1)       0.09      1.62 r
52 Multi_1/mult_16/U446/ZN (INV_X1)          0.03      1.65 f
53 Multi_1/mult_16/U668/ZN (AOI222_X1)       0.09      1.74 r
54 Multi_1/mult_16/U444/ZN (INV_X1)          0.03      1.77 f
55 Multi_1/mult_16/U667/ZN (AOI222_X1)       0.09      1.86 r
56 Multi_1/mult_16/U445/ZN (INV_X1)          0.03      1.89 f
57 Multi_1/mult_16/U666/ZN (AOI222_X1)       0.09      1.98 r
58 Multi_1/mult_16/U447/ZN (INV_X1)          0.03      2.01 f
59 Multi_1/mult_16/U665/ZN (AOI222_X1)       0.11      2.12 r
60 Multi_1/mult_16/U664/ZN (AOI222_X1)       0.07      2.18 f
61 Multi_1/mult_16/U663/ZN (AOI222_X1)       0.11      2.30 r
62 Multi_1/mult_16/U662/ZN (AOI222_X1)       0.07      2.36 f
63 Multi_1/mult_16/U13/CO (FA_X1)            0.10      2.46 f
64 Multi_1/mult_16/U12/S (FA_X1)             0.13      2.59 r
65 Multi_1/mult_16/product[18] (multiplier_N13_2_DW_mult_tc_0_DW_mult_tc_1)
66                                     0.00      2.59 r
67 Multi_1/o[18] (multiplier_N13_2)          0.00      2.59 r
68 U15/ZN (INV_X1)                          0.02      2.61 f
69 U28/Z (XOR2_X1)                          0.07      2.68 f
70 Adder1/b[6] (adder_N13_1)                0.00      2.68 f
71 Adder1/add_16/B[6] (adder_N13_1_DW01_add_0_DW01_add_1)
72                                     0.00      2.68 f
73 Adder1/add_16/U1_6/S (FA_X1)             0.19      2.87 r
74 Adder1/add_16/SUM[6] (adder_N13_1_DW01_add_0_DW01_add_1)
75                                     0.00      2.87 r
76 Adder1/o[6] (adder_N13_1)                0.00      2.87 r
77 Multi_2/b[6] (multiplier_N13_1)          0.00      2.87 r
78 Multi_2/mult_16/b[6] (multiplier_N13_1_DW_mult_tc_0_DW_mult_tc_2)
79                                     0.00      2.87 r
80 Multi_2/mult_16/U472/ZN (XNOR2_X1)        0.08      2.95 r
81 Multi_2/mult_16/U469/ZN (AOI222_X1)       0.04      3.00 f
82 Multi_2/mult_16/U104/S (FA_X1)            0.15      3.14 r
83 Multi_2/mult_16/U103/S (FA_X1)            0.11      3.26 f
84 Multi_2/mult_16/U666/ZN (AOI222_X1)       0.11      3.37 r

```

85	Multi_2/mult_16/U443/ZN (INV_X1)	0.03	3.40 f
86	Multi_2/mult_16/U665/ZN (AOI222_X1)	0.09	3.49 r
87	Multi_2/mult_16/U441/ZN (INV_X1)	0.03	3.52 f
88	Multi_2/mult_16/U664/ZN (AOI222_X1)	0.09	3.61 r
89	Multi_2/mult_16/U442/ZN (INV_X1)	0.03	3.64 f
90	Multi_2/mult_16/U663/ZN (AOI222_X1)	0.09	3.73 r
91	Multi_2/mult_16/U438/ZN (INV_X1)	0.03	3.76 f
92	Multi_2/mult_16/U662/ZN (AOI222_X1)	0.09	3.85 r
93	Multi_2/mult_16/U440/ZN (INV_X1)	0.03	3.88 f
94	Multi_2/mult_16/U661/ZN (AOI222_X1)	0.09	3.97 r
95	Multi_2/mult_16/U439/ZN (INV_X1)	0.03	4.00 f
96	Multi_2/mult_16/U660/ZN (AOI222_X1)	0.09	4.09 r
97	Multi_2/mult_16/U444/ZN (INV_X1)	0.03	4.12 f
98	Multi_2/mult_16/U659/ZN (AOI222_X1)	0.11	4.23 r
99	Multi_2/mult_16/U658/ZN (AOI222_X1)	0.07	4.29 f
100	Multi_2/mult_16/U657/ZN (AOI222_X1)	0.11	4.41 r
101	Multi_2/mult_16/U656/ZN (AOI222_X1)	0.07	4.47 f
102	Multi_2/mult_16/U13/CO (FA_X1)	0.10	4.57 f
103	Multi_2/mult_16/U12/CO (FA_X1)	0.09	4.66 f
104	Multi_2/mult_16/U11/CO (FA_X1)	0.09	4.75 f
105	Multi_2/mult_16/U10/CO (FA_X1)	0.09	4.84 f
106	Multi_2/mult_16/U9/CO (FA_X1)	0.09	4.93 f
107	Multi_2/mult_16/U8/CO (FA_X1)	0.09	5.02 f
108	Multi_2/mult_16/U7/CO (FA_X1)	0.09	5.11 f
109	Multi_2/mult_16/U6/S (FA_X1)	0.14	5.25 r
110	Multi_2/mult_16/product[24] (multiplier_N13_1_DW_mult_tc_0_DW_mult_tc_2)		
111		0.00	5.25 r
112	Multi_2/o[24] (multiplier_N13_1)	0.00	5.25 r
113	Adder2/a[12] (adder_N13_0)	0.00	5.25 r
114	Adder2/add_16/A[12] (adder_N13_0_DW01_add_0)	0.00	5.25 r
115	Adder2/add_16/U1_12/S (FA_X1)	0.11	5.36 f
116	Adder2/add_16/SUM[12] (adder_N13_0_DW01_add_0)	0.00	5.36 f
117	Adder2/o[12] (adder_N13_0)	0.00	5.36 f
118	Reg_output/D[12] (generic_reg_N13_0)	0.00	5.36 f
119	Reg_output/Q_reg[12]/D (DFFR_X1)	0.01	5.37 f
120	data arrival time		5.37
121			
122	clock MY_CLK (rise edge)	5.48	5.48
123	clock network delay (ideal)	0.00	5.48
124	clock uncertainty	-0.07	5.41
125	Reg_output/Q_reg[12]/CK (DFFR_X1)	0.00	5.41 r
126	library setup time	-0.04	5.37
127	data required time		5.37
128	-----		
129	data required time		5.37
130	data arrival time		-5.37
131	-----		
132	slack (MET)		0.00
133			
134			
135	1		

4.13 power_CLK_GATE.txt

```
1
2 *****
3 Report : power
4       -analysis_effort low
5 Design : fiir
6 Version: S-2021.06-SP4
7 Date   : Mon Oct 28 17:09:27 2024
8 *****
9
10
11 Library(s) Used:
12
13     NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
14
15
16 Operating Conditions: typical   Library: NangateOpenCellLibrary
17 Wire Load Model Mode: top
18
19 Design      Wire Load Model      Library
20 -----
21 fiir        5K_hvratio_1_1      NangateOpenCellLibrary
22
23
24 Global Operating Voltage = 1.1
25 Power-specific unit information :
26     Voltage Units = 1V
27     Capacitance Units = 1.000000ff
28     Time Units = 1ns
29     Dynamic Power Units = 1uW      (derived from V,C,T units)
30     Leakage Power Units = 1nW
31
32
33     Cell Internal Power = 109.5303 uW   (59%)
34     Net Switching Power = 76.0087 uW   (41%)
35     -----
36 Total Dynamic Power    = 185.5390 uW   (100%)
37
38 Cell Leakage Power     = 49.4190 uW
39
40
41
42 Power Group      Internal      Switching      Leakage      Total
43                  Power         Power          Power        Power ( % ) Attrs
44 -----
45 io_pad           0.0000         0.0000         0.0000        0.0000 ( 0.00%)
46 memory          0.0000         0.0000         0.0000        0.0000 ( 0.00%)
47 black_box       0.0000         0.0000         0.0000        0.0000 ( 0.00%)
48 clock_network   3.5885         3.6094        175.1728       7.3731 ( 3.14%)
49 register       18.8523        9.0408        3.6868e+03     31.5800 ( 13.44%)
50 sequential      0.0000         0.0000         0.0000        0.0000 ( 0.00%)
51 combinational   87.0893        63.3584       4.5557e+04     196.0048 ( 83.42%)
52 -----
53 Total           109.5301 uW      76.0087 uW      4.9419e+04 nW  234.9579 uW
54
55 *****
56 Report : power
57       -analysis_effort low
58 Design : fiir
59 Version: S-2021.06-SP4
60 Date   : Sat Nov 16 20:44:31 2024
61 *****
62
63
64 Library(s) Used:
65
66     NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
67
68
69 Operating Conditions: typical   Library: NangateOpenCellLibrary
70 Wire Load Model Mode: top
71
72 Design      Wire Load Model      Library
73 -----
74 fiir        5K_hvratio_1_1      NangateOpenCellLibrary
75
76
77 Global Operating Voltage = 1.1
78 Power-specific unit information :
79     Voltage Units = 1V
80     Capacitance Units = 1.000000ff
81     Time Units = 1ns
82     Dynamic Power Units = 1uW      (derived from V,C,T units)
83     Leakage Power Units = 1nW
84
```



```

85
86 Cell Internal Power = 274.1981 uW (59%)
87 Net Switching Power = 190.2619 uW (41%)
88 -----
89 Total Dynamic Power = 464.4600 uW (100%)
90
91 Cell Leakage Power = 49.4190 uW
92
93
94
95 Power Group      Internal      Switching      Leakage      Total
96 -----
97 io_pad           0.0000         0.0000         0.0000         0.0000 ( 0.00%)
98 memory           0.0000         0.0000         0.0000         0.0000 ( 0.00%)
99 black_box        0.0000         0.0000         0.0000         0.0000 ( 0.00%)
100 clock_network    8.9858         8.9181         175.2875        18.0792 ( 3.52%)
101 register         46.7531        22.4628        3.6906e+03       72.9065 ( 14.19%)
102 sequential       0.0000         0.0000         0.0000         0.0000 ( 0.00%)
103 combinational    218.4594       158.8809       4.5553e+04       422.8928 ( 82.29%)
104 -----
105 Total            274.1982 uW    190.2618 uW    4.9419e+04 nW    513.8785 uW
106 1

```

4.14 timing_report_innovus.txt

```
1 #####
2 # Generated by: Cadence Innovus 20.11-s130_1
3 # OS: Linux x86_64(Host ID isaserver)
4 # Generated on: Mon Nov 18 22:12:23 2024
5 # Design: fiir
6 # Command: report_timing >> timing_report.txt
7 #####
8 Path 1: MET Setup Check with Pin Reg_output_Q_reg_l2_/CK
9 Endpoint: Reg_output_Q_reg_l2_/D (v) checked with leading edge of 'MY_CLK'
10 Beginpoint: al[0] (v) triggered by leading edge of 'MY_CLK'
11 Path Groups: {MY_CLK}
12 Analysis View: MyAnView
13 Other End Arrival Time 0.000
14 - Setup 0.042
15 + Phase Shift 10.960
16 - Uncertainty 0.070
17 = Required Time 10.848
18 - Arrival Time 4.388
19 = Slack Time 6.460
20 Clock Rise Edge 0.000
21 + Input Delay 0.500
22 = Beginpoint Arrival Time 0.500
23 +-----+
24 | Instance | Arc | Cell | Delay | Arrival | Required |
25 | | | | | Time | Time |
26 |-----|
27 | | al[0] v | | | 0.500 | 6.960 |
28 | Multi_1_mult_16_U423 | A v -> ZN ^ | INV_X1 | 0.069 | 0.569 | 7.029 |
29 | Multi_1_mult_16_U454 | A2 ^ -> ZN v | NAND2_X1 | 0.073 | 0.643 | 7.103 |
30 | Multi_1_mult_16_U469 | A2 v -> ZN ^ | OAI22_X1 | 0.060 | 0.702 | 7.162 |
31 | Multi_1_mult_16_U107 | A ^ -> S v | FA_X1 | 0.100 | 0.802 | 7.263 |
32 | Multi_1_mult_16_U674 | C2 v -> ZN ^ | AOI222_X1 | 0.097 | 0.899 | 7.359 |
33 | Multi_1_mult_16_U450 | A ^ -> ZN v | INV_X1 | 0.014 | 0.913 | 7.374 |
34 | Multi_1_mult_16_U673 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 0.991 | 7.451 |
35 | Multi_1_mult_16_U448 | A ^ -> ZN v | INV_X1 | 0.014 | 1.005 | 7.466 |
36 | Multi_1_mult_16_U672 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 1.083 | 7.544 |
37 | Multi_1_mult_16_U449 | A ^ -> ZN v | INV_X1 | 0.015 | 1.098 | 7.559 |
38 | Multi_1_mult_16_U671 | B1 v -> ZN ^ | AOI222_X1 | 0.082 | 1.180 | 7.640 |
39 | Multi_1_mult_16_U434 | A ^ -> ZN v | INV_X1 | 0.015 | 1.194 | 7.655 |
40 | Multi_1_mult_16_U670 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.271 | 7.732 |
41 | Multi_1_mult_16_U443 | A ^ -> ZN v | INV_X1 | 0.014 | 1.286 | 7.746 |
42 | Multi_1_mult_16_U669 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.363 | 7.823 |
43 | Multi_1_mult_16_U446 | A ^ -> ZN v | INV_X1 | 0.015 | 1.378 | 7.839 |
44 | Multi_1_mult_16_U668 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.455 | 7.916 |
45 | Multi_1_mult_16_U444 | A ^ -> ZN v | INV_X1 | 0.014 | 1.470 | 7.930 |
46 | Multi_1_mult_16_U667 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 1.548 | 8.008 |
47 | Multi_1_mult_16_U445 | A ^ -> ZN v | INV_X1 | 0.016 | 1.563 | 8.024 |
48 | Multi_1_mult_16_U666 | B1 v -> ZN ^ | AOI222_X1 | 0.079 | 1.643 | 8.103 |
49 | Multi_1_mult_16_U447 | A ^ -> ZN v | INV_X1 | 0.014 | 1.657 | 8.117 |
50 | Multi_1_mult_16_U665 | B1 v -> ZN ^ | AOI222_X1 | 0.093 | 1.750 | 8.210 |
51 | Multi_1_mult_16_U664 | B1 ^ -> ZN v | OAI222_X1 | 0.054 | 1.803 | 8.263 |
52 | Multi_1_mult_16_U663 | B1 v -> ZN ^ | AOI222_X1 | 0.098 | 1.901 | 8.362 |
53 | Multi_1_mult_16_U662 | B1 ^ -> ZN v | OAI222_X1 | 0.055 | 1.956 | 8.416 |
54 | Multi_1_mult_16_U13 | CI v -> CO v | FA_X1 | 0.078 | 2.034 | 8.495 |
55 | Multi_1_mult_16_U12 | CI v -> CO v | FA_X1 | 0.072 | 2.106 | 8.566 |
56 | Multi_1_mult_16_U11 | CI v -> S ^ | FA_X1 | 0.112 | 2.218 | 8.678 |
57 | U7 | A ^ -> ZN v | INV_X1 | 0.011 | 2.229 | 8.689 |
58 | U26 | A v -> Z v | XOR2_X1 | 0.053 | 2.282 | 8.742 |
59 | Adder1_add_16_U1_7 | B v -> CO v | FA_X1 | 0.082 | 2.364 | 8.824 |
60 | Adder1_add_16_U1_8 | CI v -> CO v | FA_X1 | 0.073 | 2.437 | 8.897 |
61 | Adder1_add_16_U1_9 | CI v -> CO v | FA_X1 | 0.072 | 2.510 | 8.970 |
62 | Adder1_add_16_U1_10 | CI v -> S ^ | FA_X1 | 0.163 | 2.673 | 9.133 |
63 | Multi_2_mult_16_U480 | A ^ -> ZN v | XNOR2_X1 | 0.030 | 2.703 | 9.163 |
64 | Multi_2_mult_16_U477 | B1 v -> ZN ^ | OAI22_X1 | 0.055 | 2.757 | 9.218 |
65 | Multi_2_mult_16_U92 | CI ^ -> S v | FA_X1 | 0.104 | 2.861 | 9.321 |
66 | Multi_2_mult_16_U90 | CI v -> S ^ | FA_X1 | 0.116 | 2.977 | 9.437 |
67 | Multi_2_mult_16_U89 | CI ^ -> S v | FA_X1 | 0.093 | 3.070 | 9.530 |
68 | Multi_2_mult_16_U662 | C2 v -> ZN ^ | AOI222_X1 | 0.098 | 3.168 | 9.628 |
69 | Multi_2_mult_16_U440 | A ^ -> ZN v | INV_X1 | 0.015 | 3.183 | 9.643 |
70 | Multi_2_mult_16_U661 | B1 v -> ZN ^ | AOI222_X1 | 0.079 | 3.262 | 9.722 |
71 | Multi_2_mult_16_U439 | A ^ -> ZN v | INV_X1 | 0.014 | 3.276 | 9.736 |
72 | Multi_2_mult_16_U660 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 3.354 | 9.814 |
73 | Multi_2_mult_16_U444 | A ^ -> ZN v | INV_X1 | 0.014 | 3.368 | 9.828 |
74 | Multi_2_mult_16_U659 | B1 v -> ZN ^ | AOI222_X1 | 0.096 | 3.464 | 9.924 |
75 | Multi_2_mult_16_U658 | B1 ^ -> ZN v | OAI222_X1 | 0.054 | 3.517 | 9.978 |
76 | Multi_2_mult_16_U657 | B1 v -> ZN ^ | AOI222_X1 | 0.098 | 3.615 | 10.075 |
77 | Multi_2_mult_16_U656 | B1 ^ -> ZN v | OAI222_X1 | 0.051 | 3.667 | 10.127 |
78 | Multi_2_mult_16_U13 | CI v -> CO v | FA_X1 | 0.077 | 3.744 | 10.204 |
79 | Multi_2_mult_16_U12 | CI v -> CO v | FA_X1 | 0.075 | 3.819 | 10.280 |
80 | Multi_2_mult_16_U11 | CI v -> CO v | FA_X1 | 0.074 | 3.893 | 10.354 |
81 | Multi_2_mult_16_U10 | CI v -> CO v | FA_X1 | 0.072 | 3.965 | 10.426 |
82 | Multi_2_mult_16_U9 | CI v -> CO v | FA_X1 | 0.071 | 4.037 | 10.497 |
83 | Multi_2_mult_16_U8 | CI v -> CO v | FA_X1 | 0.071 | 4.108 | 10.568 |
84 | Multi_2_mult_16_U7 | CI v -> CO v | FA_X1 | 0.072 | 4.180 | 10.640 |
```

```

85 | Multi_2_mult_16_U6 | CI v -> S ^ | FA_X1 | 0.121 | 4.301 | 10.761 |
86 | Adder2_add_16_U1_12 | A ^ -> S v | FA_X1 | 0.087 | 4.388 | 10.848 |
87 | Reg_output_Q_reg_12_ | D v | DFFR_X1 | 0.000 | 4.388 | 10.848 |
88 +-----+
89
90 #####
91 # Generated by: Cadence Innovus 20.11-s130.1
92 # OS: Linux x86_64(Host ID isaserver)
93 # Generated on: Mon Nov 18 22:12:41 2024
94 # Design: fiir
95 # Command: report_timing >> timing_report.txt
96 #####
97 Path 1: MET Setup Check with Pin Reg_output_Q_reg_12_/CK
98 Endpoint: Reg_output_Q_reg_12_/D (v) checked with leading edge of 'MY_CLK'
99 Beginpoint: a1[0] (v) triggered by leading edge of 'MY_CLK'
100 Path Groups: {MY_CLK}
101 Analysis View: MyAnView
102 Other End Arrival Time 0.000
103 - Setup 0.042
104 + Phase Shift 10.960
105 - Uncertainty 0.070
106 = Required Time 10.848
107 - Arrival Time 4.388
108 = Slack Time 6.460
109 Clock Rise Edge 0.000
110 + Input Delay 0.500
111 = Beginpoint Arrival Time 0.500
112 +-----+
113 | Instance | Arc | Cell | Delay | Arrival | Required |
114 | | | | | Time | Time |
115 +-----+
116 | Multi_1_mult_16_U423 | a1[0] v | INV_X1 | 0.069 | 0.500 | 6.960 |
117 | Multi_1_mult_16_U454 | A2 ^ -> ZN ^ | NAND2_X1 | 0.073 | 0.569 | 7.029 |
118 | Multi_1_mult_16_U469 | A2 v -> ZN ^ | OAI222_X1 | 0.060 | 0.643 | 7.103 |
119 | Multi_1_mult_16_U107 | A ^ -> S v | FA_X1 | 0.100 | 0.702 | 7.162 |
120 | Multi_1_mult_16_U674 | C2 v -> ZN ^ | AOI222_X1 | 0.097 | 0.802 | 7.263 |
121 | Multi_1_mult_16_U450 | A ^ -> ZN v | INV_X1 | 0.097 | 0.899 | 7.359 |
122 | Multi_1_mult_16_U673 | B1 v -> ZN ^ | INV_X1 | 0.014 | 0.913 | 7.374 |
123 | Multi_1_mult_16_U448 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 0.991 | 7.451 |
124 | Multi_1_mult_16_U672 | A ^ -> ZN v | INV_X1 | 0.014 | 1.005 | 7.466 |
125 | Multi_1_mult_16_U449 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 1.083 | 7.544 |
126 | Multi_1_mult_16_U671 | A ^ -> ZN v | INV_X1 | 0.015 | 1.098 | 7.559 |
127 | Multi_1_mult_16_U671 | B1 v -> ZN ^ | AOI222_X1 | 0.082 | 1.180 | 7.640 |
128 | Multi_1_mult_16_U434 | A ^ -> ZN v | INV_X1 | 0.015 | 1.194 | 7.655 |
129 | Multi_1_mult_16_U670 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.271 | 7.732 |
130 | Multi_1_mult_16_U443 | A ^ -> ZN v | INV_X1 | 0.014 | 1.286 | 7.746 |
131 | Multi_1_mult_16_U669 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.363 | 7.823 |
132 | Multi_1_mult_16_U446 | A ^ -> ZN v | INV_X1 | 0.015 | 1.378 | 7.839 |
133 | Multi_1_mult_16_U668 | B1 v -> ZN ^ | AOI222_X1 | 0.077 | 1.455 | 7.916 |
134 | Multi_1_mult_16_U444 | A ^ -> ZN v | INV_X1 | 0.014 | 1.470 | 7.930 |
135 | Multi_1_mult_16_U667 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 1.548 | 8.008 |
136 | Multi_1_mult_16_U445 | A ^ -> ZN v | INV_X1 | 0.016 | 1.563 | 8.024 |
137 | Multi_1_mult_16_U666 | B1 v -> ZN ^ | AOI222_X1 | 0.079 | 1.643 | 8.103 |
138 | Multi_1_mult_16_U447 | A ^ -> ZN v | INV_X1 | 0.014 | 1.657 | 8.117 |
139 | Multi_1_mult_16_U665 | B1 v -> ZN ^ | AOI222_X1 | 0.093 | 1.750 | 8.210 |
140 | Multi_1_mult_16_U664 | B1 ^ -> ZN v | OAI222_X1 | 0.054 | 1.803 | 8.263 |
141 | Multi_1_mult_16_U663 | B1 v -> ZN ^ | AOI222_X1 | 0.098 | 1.901 | 8.362 |
142 | Multi_1_mult_16_U662 | B1 ^ -> ZN v | OAI222_X1 | 0.055 | 1.956 | 8.416 |
143 | Multi_1_mult_16_U13 | CI v -> CO v | FA_X1 | 0.078 | 2.034 | 8.495 |
144 | Multi_1_mult_16_U12 | CI v -> CO v | FA_X1 | 0.072 | 2.106 | 8.566 |
145 | Multi_1_mult_16_U11 | CI v -> S ^ | FA_X1 | 0.112 | 2.218 | 8.678 |
146 | U7 | A ^ -> ZN v | INV_X1 | 0.011 | 2.229 | 8.689 |
147 | U26 | A v -> Z v | XOR2_X1 | 0.053 | 2.282 | 8.742 |
148 | Adder1_add_16_U1_7 | B v -> CO v | FA_X1 | 0.082 | 2.364 | 8.824 |
149 | Adder1_add_16_U1_8 | CI v -> CO v | FA_X1 | 0.073 | 2.437 | 8.897 |
150 | Adder1_add_16_U1_9 | CI v -> CO v | FA_X1 | 0.072 | 2.510 | 8.970 |
151 | Adder1_add_16_U1_10 | CI v -> S ^ | FA_X1 | 0.163 | 2.673 | 9.133 |
152 | Multi_2_mult_16_U480 | A ^ -> ZN v | XNOR2_X1 | 0.030 | 2.703 | 9.163 |
153 | Multi_2_mult_16_U477 | B1 v -> ZN ^ | OAI222_X1 | 0.055 | 2.757 | 9.218 |
154 | Multi_2_mult_16_U92 | CI ^ -> S v | FA_X1 | 0.104 | 2.861 | 9.321 |
155 | Multi_2_mult_16_U90 | CI v -> S ^ | FA_X1 | 0.116 | 2.977 | 9.437 |
156 | Multi_2_mult_16_U89 | CI ^ -> S v | FA_X1 | 0.093 | 3.070 | 9.530 |
157 | Multi_2_mult_16_U662 | C2 v -> ZN ^ | AOI222_X1 | 0.098 | 3.168 | 9.628 |
158 | Multi_2_mult_16_U440 | A ^ -> ZN v | INV_X1 | 0.015 | 3.183 | 9.643 |
159 | Multi_2_mult_16_U661 | B1 v -> ZN ^ | AOI222_X1 | 0.079 | 3.262 | 9.722 |
160 | Multi_2_mult_16_U439 | A ^ -> ZN v | INV_X1 | 0.014 | 3.276 | 9.736 |
161 | Multi_2_mult_16_U660 | B1 v -> ZN ^ | AOI222_X1 | 0.078 | 3.354 | 9.814 |
162 | Multi_2_mult_16_U444 | A ^ -> ZN v | INV_X1 | 0.014 | 3.368 | 9.828 |
163 | Multi_2_mult_16_U659 | B1 v -> ZN ^ | AOI222_X1 | 0.096 | 3.464 | 9.924 |
164 | Multi_2_mult_16_U658 | B1 ^ -> ZN v | OAI222_X1 | 0.054 | 3.517 | 9.978 |
165 | Multi_2_mult_16_U657 | B1 v -> ZN ^ | AOI222_X1 | 0.098 | 3.615 | 10.075 |
166 | Multi_2_mult_16_U656 | B1 ^ -> ZN v | OAI222_X1 | 0.051 | 3.667 | 10.127 |
167 | Multi_2_mult_16_U13 | CI v -> CO v | FA_X1 | 0.077 | 3.744 | 10.204 |
168 | Multi_2_mult_16_U12 | CI v -> CO v | FA_X1 | 0.075 | 3.819 | 10.280 |
169 | Multi_2_mult_16_U11 | CI v -> CO v | FA_X1 | 0.074 | 3.893 | 10.354 |
170 | Multi_2_mult_16_U10 | CI v -> CO v | FA_X1 | 0.072 | 3.965 | 10.426 |
171 | Multi_2_mult_16_U9 | CI v -> CO v | FA_X1 | 0.071 | 4.037 | 10.497 |
172 | Multi_2_mult_16_U8 | CI v -> CO v | FA_X1 | 0.071 | 4.108 | 10.568 |
173 | Multi_2_mult_16_U7 | CI v -> CO v | FA_X1 | 0.072 | 4.180 | 10.640 |
174 | Multi_2_mult_16_U6 | CI v -> S ^ | FA_X1 | 0.121 | 4.301 | 10.761 |
175 | Adder2_add_16_U1_12 | A ^ -> S v | FA_X1 | 0.087 | 4.388 | 10.848 |
176 | Reg_output_Q_reg_12_ | D v | DFFR_X1 | 0.000 | 4.388 | 10.848 |
177 +-----+

```


4.15 area_report_innovus.txt

1	Hinst Name	Module Name	Inst Count	Total Area
2	-----	-----	-----	-----
3	fiir		1173	2467.682
4	Mid_reg_clk_gate_Q_reg	SNPS_CLOCK_GATE_HIGH_generic_reg_N14	1	7.714
5	Reg_input_clk_gate_Q_reg	SNPS_CLOCK_GATE_HIGH_generic_reg_N13_0_1	1	7.714
6	Reg_output_clk_gate_Q_reg	SNPS_CLOCK_GATE_HIGH_generic_reg_N13_0_0	1	7.714

4.16 myfir_advanced.c

```
1 #include<stdio.h>
2 #include<stdlib.h>
3
4 #define N 2 /// order of the filter
5 #define NTm1 (N) /// number of coefficients minus one (equal to the order)
6 #define NB 13 /// number of bits
7 #define SHAMT 18 /// shift amount
8
9 const int bi0 = 1723;          /// coefficient b0 also b1=b0=1723 , a1=-649
10 const int bi[NTm1]={1996,273}; /// b array: bi0=b1-a1*b0 , b1=-a1*b1
11 const int ai[NTm1]={0,102}; /// a array: ai0=0 , a1=a1*a1
12
13 /// Perform lookahead on the IIR filter
14 ///\param x is the new input sample
15 ///\return the new output sample
16 int myfilter(int x)
17 {
18     static int sw[NTm1]; /// w shift register
19     static int first_run = 0; /// for cleaning the shift register
20     int i; /// index
21     int w; /// intermediate value (w)
22     int y; /// output sample
23     int fb, ff; /// feed-back and feed-forward results
24
25     /// clean the buffer
26     if (first_run == 0)
27     {
28         first_run = 1;
29         for (i=0; i<NTm1; i++)
30             sw[i] = 0;
31     }
32
33     /// compute feed-back and feed-forward
34     fb = 0;
35     ff = 0;
36     for (i=0; i<NTm1; i++)
37     {
38         fb += ((sw[i]*ai[i]) >> SHAMT) << (SHAMT-NB+1);
39         ff += ((sw[i]*bi[i]) >> SHAMT) << (SHAMT-NB+1);
40     }
41
42     /// compute intermediate value (w) and output sample
43     w = x + fb;
44     y = (w*bi0) >> SHAMT << (SHAMT-NB+1);
45     y += ff;
46
47     /// update the shift register
48     for (i=NTm1; i>0; i--)
49         sw[i] = sw[i-1];
50     sw[0] = w;
51
52     return y;
53 }
54
55 int main (int argc, char **argv)
56 {
57     FILE *fp_in;
58     FILE *fp_out;
59
60     int x;
61     int y;
62
63     /// check the command line
64     if (argc != 3)
65     {
66         printf("Use: %s <input_file> <output_file>\n", argv[0]);
67         exit(1);
68     }
69
70     /// open files
71     fp_in = fopen(argv[1], "r");
72     if (fp_in == NULL)
73     {
74         printf("Error: cannot open %s\n", argv[1]);
75         exit(2);
76     }
77     fp_out = fopen(argv[2], "w");
78
79     /// check shift amount
80     if (SHAMT < ((NB)-1))
81     {
82         printf("Error shift amount must be at least nbit-1\n");
83         exit(3);
84     }
```

```
85
86  /// get samples and apply filter
87  fscanf(fp_in, "%d", &x);
88  do{
89      y = myfilter(x);
90      fprintf(fp_out, "%d\n", y);
91      fscanf(fp_in, "%d", &x);
92  } while (!feof(fp_in));
93
94  fclose(fp_in);
95  fclose(fp_out);
96
97  return 0;
98
99 }
```

4.17 resultc_18_lookahead.txt

```
1 0
2 512
3 576
4 1408
5 1600
6 1920
7 1920
8 1600
9 1472
10 640
11 512
12 -576
13 -768
14 -1664
15 -1792
16 -2112
17 -2240
18 -1856
19 -1856
20 -960
21 -832
22 256
23 448
24 1344
25 1472
26 1856
27 1856
28 1600
29 1472
30 640
31 512
32 -576
33 -768
34 -1664
35 -1792
36 -2112
37 -2240
38 -1856
39 -1856
40 -960
41 -832
42 256
43 448
44 1344
45 1472
46 1856
47 1856
48 1600
49 1472
50 640
51 512
52 -576
53 -768
54 -1664
55 -1792
56 -2112
57 -2240
58 -1856
59 -1856
60 -960
61 -832
62 256
63 448
64 1344
65 1472
66 1856
67 1856
68 1600
69 1472
70 640
71 512
72 -576
73 -768
74 -1664
75 -1792
76 -2112
77 -2240
78 -1856
79 -1856
80 -960
81 -832
82 256
83 448
84 1344
```


85 1472
86 1856
87 1856
88 1600
89 1472
90 640
91 512
92 -576
93 -768
94 -1664
95 -1792
96 -2112
97 -2240
98 -1856
99 -1856
100 -960
101 -832
102 256
103 448
104 1344
105 1472
106 1856
107 1856
108 1600
109 1472
110 640
111 512
112 -576
113 -768
114 -1664
115 -1792
116 -2112
117 -2240
118 -1856
119 -1856
120 -960
121 -832
122 256
123 448
124 1344
125 1472
126 1856
127 1856
128 1600
129 1472
130 640
131 512
132 -576
133 -768
134 -1664
135 -1792
136 -2112
137 -2240
138 -1856
139 -1856
140 -960
141 -832
142 256
143 448
144 1344
145 1472
146 1856
147 1856
148 1600
149 1472
150 640
151 512
152 -576
153 -768
154 -1664
155 -1792
156 -2112
157 -2240
158 -1856
159 -1856
160 -960
161 -832
162 256
163 448
164 1344
165 1472
166 1856
167 1856
168 1600
169 1472
170 640
171 512
172 -576
173 -768
174 -1664
175 -1792
176 -2112
177 -2240
178 -1856
179 -1856

180	-960
181	-832
182	256
183	448
184	1344
185	1472
186	1856
187	1856
188	1600
189	1472
190	640
191	512
192	-576
193	-768
194	-1664
195	-1792
196	-2112
197	-2240
198	-1856
199	-1856
200	-960
201	-832

4.18 ad_area_without_clk_gate.txt

```
1
2 *****
3 Report : area
4 Design : fiir_lookahead
5 Version: S-2021.06-SP4
6 Date   : Fri Nov 22 20:55:29 2024
7 *****
8
9 Library(s) Used:
10
11   NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
12
13 Number of ports:          1125
14 Number of nets:          4035
15 Number of cells:         2659
16 Number of combinational cells: 2470
17 Number of sequential cells:   156
18 Number of macros/black boxes:    0
19 Number of buf/inv:          319
20 Number of references:        23
21
22 Combinational area:        3708.306001
23 Buf/Inv area:              194.445999
24 Noncombinational area:     830.452027
25 Macro/Black Box area:      0.000000
26 Net Interconnect area:     undefined (Wire load has zero net area)
27
28 Total cell area:           4538.758027
29 Total area:                undefined
30 1
```

4.19 ad_power_without_clk_gate.txt

```
1 Information: Updating design information... (UID-85)
2 Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
3 Warning: The derived toggle rate value (1.111111) for the clock net 'CLK' conflicts with the annotated value (1.000000). Using the annotated value
4
5 *****
6 Report : power
7         -analysis_effort low
8 Design : fiir_lookahead
9 Version: S-2021.06-SP4
10 Date   : Sat Nov 23 21:48:28 2024
11 *****
12
13
14 Library(s) Used:
15
16     NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
17
18
19 Operating Conditions: typical    Library: NangateOpenCellLibrary
20 Wire Load Model Mode: top
21
22 Design      Wire Load Model      Library
23 -----
24 fiir_lookahead      5K_hvratio_1_1      NangateOpenCellLibrary
25
26
27 Global Operating Voltage = 1.1
28 Power-specific unit information :
29     Voltage Units = 1V
30     Capacitance Units = 1.000000ff
31     Time Units = 1ns
32     Dynamic Power Units = 1uW      (derived from V,C,T units)
33     Leakage Power Units = 1nW
34
35
36 Cell Internal Power = 1.2373 mW      (59%)
37 Net Switching Power = 846.0948 uW      (41%)
38 -----
39 Total Dynamic Power = 2.0834 mW      (100%)
40
41 Cell Leakage Power = 95.7553 uW
42
43
44
45 Power Group      Internal      Switching      Leakage      Total
46 -----
47 io_pad           0.0000         0.0000         0.0000         0.0000 ( 0.00%)
48 memory          0.0000         0.0000         0.0000         0.0000 ( 0.00%)
49 black_box       0.0000         0.0000         0.0000         0.0000 ( 0.00%)
50 clock_network   0.0000         0.0000         0.0000         0.0000 ( 0.00%)
51 register        525.6376       151.2249       1.3762e+04      690.6247 ( 31.69%)
52 sequential      0.0000         0.0000         0.0000         0.0000 ( 0.00%)
53 combinational   711.6884       694.8729       8.1993e+04      1.4886e+03 ( 68.31%)
54 -----
55 Total           1.2373e+03 uW      846.0979 uW      9.5755e+04 nW      2.1792e+03 uW
56 1
```

4.20 ad_timing_without_clk_gate.txt

```

1 Information: Updating design information... (UID-85)
2
3 *****
4 Report : timing
5       -path full
6       -delay max
7       -max_paths 1
8 Design : fiir_lookahead
9 Version: S-2021.06-SP4
10 Date   : Mon Nov 11 11:49:36 2024
11 *****
12
13 Operating Conditions: typical      Library: NangateOpenCellLibrary
14 Wire Load Model Mode: top
15
16 Startpoint: b1[12] (input port clocked by MY_CLK)
17 Endpoint:   Mid_reg_6/Q_reg[13]
18             (rising edge-triggered flip-flop clocked by MY_CLK)
19 Path Group: MY_CLK
20 Path Type:  max
21
22 Des/Clust/Port      Wire Load Model      Library
23 -----
24 fiir_lookahead      5K_hvratio_1_1      NangateOpenCellLibrary
25
26 Point                                     Incr      Path
27 -----
28 clock MY_CLK (rise edge)                  0.00      0.00
29 clock network delay (ideal)                0.00      0.00
30 input external delay                      0.50      0.50 f
31 b1[12] (in)                               0.00      0.50 f
32 Multi_3/a[12] (multiplier_N13_1)          0.00      0.50 f
33 Multi_3/mult_16/a[12] (multiplier_N13_1_DW_mult_tc_1)
34                                             0.00      0.50 f
35 Multi_3/mult_16/U885/ZN (XNOR2_X1)         0.06      0.56 f
36 Multi_3/mult_16/U965/ZN (OAI22_X1)         0.06      0.62 r
37 Multi_3/mult_16/U896/ZN (INV_X1)           0.03      0.65 f
38 Multi_3/mult_16/U205/CO (FA_X1)            0.09      0.74 f
39 Multi_3/mult_16/U199/CO (FA_X1)            0.10      0.84 f
40 Multi_3/mult_16/U811/ZN (NAND2_X1)         0.03      0.88 r
41 Multi_3/mult_16/U813/ZN (NAND3_X1)         0.04      0.92 f
42 Multi_3/mult_16/U188/S (FA_X1)            0.13      1.05 f
43 Multi_3/mult_16/U852/ZN (NAND2_X1)         0.04      1.09 r
44 Multi_3/mult_16/U920/ZN (OAI21_X1)         0.04      1.13 f
45 Multi_3/mult_16/U855/ZN (AOI21_X1)         0.04      1.18 r
46 Multi_3/mult_16/U932/ZN (OAI21_X1)         0.03      1.21 f
47 Multi_3/mult_16/U582/ZN (AOI21_X1)         0.06      1.27 r
48 Multi_3/mult_16/U947/ZN (OAI21_X1)         0.03      1.30 f
49 Multi_3/mult_16/U573/ZN (AOI21_X1)         0.04      1.34 r
50 Multi_3/mult_16/U950/ZN (OAI21_X1)         0.03      1.38 f
51 Multi_3/mult_16/U6/CO (FA_X1)              0.10      1.48 f
52 Multi_3/mult_16/U546/ZN (NAND2_X1)         0.04      1.51 r
53 Multi_3/mult_16/U547/ZN (NAND3_X1)         0.04      1.55 f
54 Multi_3/mult_16/U540/ZN (XNOR2_X1)         0.06      1.61 f
55 Multi_3/mult_16/product[25] (multiplier_N13_1_DW_mult_tc_1)
56                                             0.00      1.61 f
57 Multi_3/o[25] (multiplier_N13_1)           0.00      1.61 f
58 Mid_reg_6/D[13] (generic_reg_N14_2)        0.00      1.61 f
59 Mid_reg_6/U27/Z (MUX2_X1)                 0.06      1.68 f
60 Mid_reg_6/Q_reg[13]/D (DFFR_X1)           0.01      1.69 f
61 data arrival time                          1.69
62
63 clock MY_CLK (rise edge)                  1.80      1.80
64 clock network delay (ideal)                0.00      1.80
65 clock uncertainty                          -0.07      1.73
66 Mid_reg_6/Q_reg[13]/CK (DFFR_X1)          0.00      1.73 r
67 library setup time                        -0.04      1.69
68 data required time                         1.69
69 -----
70 data required time                         1.69
71 data arrival time                         -1.69
72 -----
73 slack (MET)                               0.00
74
75
76 1

```

4.21 ad_area_clk_gate.txt

```
1
2 *****
3 Report : area
4 Design : fiir_lookahead
5 Version: S-2021.06-SP4
6 Date   : Mon Nov 18 23:51:45 2024
7 *****
8
9 Library(s) Used:
10
11 gtech (File: /eda/synopsys/2021-22/RHELx86/SYN_2021.06-SP4/libraries/syn/gtech.db)
12
13 Number of ports:          1471
14 Number of nets:          4167
15 Number of cells:         1964
16 Number of combinational cells: 1737
17 Number of sequential cells:  160
18 Number of macros/black boxes: 0
19 Number of buf/inv:        123
20 Number of references:      22
21
22 Combinational area:      0.000000
23 Buf/Inv area:            0.000000
24 Noncombinational area:   0.000000
25 Macro/Black Box area:    0.000000
26 Net Interconnect area:   undefined (No wire load specified)
27
28 Total cell area:         0.000000
29 Total area:              undefined
30
31 Information: This design contains unmapped logic. (RPT-7)
32 1
33
34 *****
35 Report : area
36 Design : fiir_lookahead
37 Version: S-2021.06-SP4
38 Date   : Mon Nov 18 23:57:11 2024
39 *****
40
41 Library(s) Used:
42
43 NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
44
45 Number of ports:          1169
46 Number of nets:          3716
47 Number of cells:         2414
48 Number of combinational cells: 2192
49 Number of sequential cells:  167
50 Number of macros/black boxes: 0
51 Number of buf/inv:        265
52 Number of references:      22
53
54 Combinational area:      3404.268002
55 Buf/Inv area:            154.546000
56 Noncombinational area:   874.874027
57 Macro/Black Box area:    0.000000
58 Net Interconnect area:   undefined (Wire load has zero net area)
59
60 Total cell area:         4279.142029
61 Total area:              undefined
62 1
63
64 *****
65 Report : area
66 Design : fiir_lookahead
67 Version: S-2021.06-SP4
68 Date   : Thu Nov 21 20:56:10 2024
69 *****
70
71 Library(s) Used:
72
73 NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
74
75 Number of ports:          1169
76 Number of nets:          3716
77 Number of cells:         2414
78 Number of combinational cells: 2192
79 Number of sequential cells:  167
80 Number of macros/black boxes: 0
81 Number of buf/inv:        265
82 Number of references:      22
83
84 Combinational area:      3404.268002
```

```
85 Buf/Inv area:          154.546000
86 Noncombinational area: 874.874027
87 Macro/Black Box area:  0.000000
88 Net Interconnect area: undefined (Wire load has zero net area)
89
90 Total cell area:       4279.142029
91 Total area:            undefined
92 1
```

4.22 ad_power_clk_gate.txt

```
1 Information: Updating design information... (UID-85)
2 Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)
3 Warning: The derived toggle rate value (1.136364) for the clock net 'CLK' conflicts with the annotated value (1.000000). Using the annotated value
4
5 *****
6 Report : power
7         -analysis_effort low
8 Design : fiir_lookahead
9 Version: S-2021.06-SP4
10 Date   : Sat Nov 23 20:45:08 2024
11 *****
12
13
14 Library(s) Used:
15
16     NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/NangateOpenCellLibrary_typical_ecsm.db)
17
18
19 Operating Conditions: typical    Library: NangateOpenCellLibrary
20 Wire Load Model Mode: top
21
22 Design      Wire Load Model      Library
23 -----
24 fiir_lookahead      5K_hvratio_1_1      NangateOpenCellLibrary
25
26
27 Global Operating Voltage = 1.1
28 Power-specific unit information :
29     Voltage Units = 1V
30     Capacitance Units = 1.000000ff
31     Time Units = 1ns
32     Dynamic Power Units = 1uW      (derived from V,C,T units)
33     Leakage Power Units = 1nW
34
35
36 Cell Internal Power = 1.0729 mW (57%)
37 Net Switching Power = 803.6547 uW (43%)
38 -----
39 Total Dynamic Power = 1.8765 mW (100%)
40
41 Cell Leakage Power = 88.7096 uW
42
43
44
45 Power Group      Internal      Switching      Leakage      Total
46 -----      Power      Power      Power      Power ( % ) Attrs
47 io_pad           0.0000           0.0000           0.0000           0.0000 ( 0.00%)
48 memory           0.0000           0.0000           0.0000           0.0000 ( 0.00%)
49 black_box        0.0000           0.0000           0.0000           0.0000 ( 0.00%)
50 clock_network    66.5117          69.9286          641.9340         137.0822 ( 6.98%)
51 register         358.6465         143.5507         1.3809e+04        516.0062 ( 26.26%)
52 sequential       0.0000           0.0000           0.0000           0.0000 ( 0.00%)
53 combinational    647.7198         590.1756         7.4259e+04        1.3122e+03 ( 66.77%)
54 -----
55 Total           1.0729e+03 uW      803.6549 uW      8.8710e+04 nW      1.9652e+03 uW
56 1
```


4.23 ad_timing_clk_gate.txt

```

1
2
3 *****
4 Report : timing
5     -path full
6     -delay max
7     -max_paths 1
8 Design : fiir_lookahead
9 Version: S-2021.06-SP4
10 Date   : Mon Nov 18 23:51:45 2024
11 *****
12
13 Operating Conditions: nom_pvt   Library: gtech
14 Wire Load Model Mode: top
15
16 Startpoint: VIN (input port clocked by MY_CLK)
17 Endpoint: FF1/Q_reg (rising edge-triggered flip-flop clocked by MY_CLK)
18 Path Group: MY_CLK
19 Path Type: max
20
21 Point                                     Incr      Path
22 -----
23 clock MY_CLK (rise edge)                  0.00      0.00
24 clock network delay (ideal)                0.00      0.00
25 input external delay                      0.50      0.50 r
26 VIN (in)                                  0.00      0.50 r
27 FF1/Din (DFF_syn_3)                       0.00      0.50 r
28 FF1/Q_reg/next_state (**SEQGEN**)          0.00      0.50 r
29 data arrival time                          0.50
30
31 clock MY_CLK (rise edge)                  1.76      1.76
32 clock network delay (ideal)                0.00      1.76
33 clock uncertainty                         -0.07      1.69
34 FF1/Q_reg/clocked_on (**SEQGEN**)          0.00      1.69 r
35 library setup time                        0.00      1.69
36 data required time                        1.69
37 -----
38 data required time                        1.69
39 data arrival time                         -0.50
40 -----
41 slack (MET)                              1.19
42
43
44 1
45 Information: Updating design information... (UID=85)
46
47 *****
48 Report : timing
49     -path full
50     -delay max
51     -max_paths 1
52 Design : fiir_lookahead
53 Version: S-2021.06-SP4
54 Date   : Mon Nov 18 23:57:11 2024
55 *****
56
57 Operating Conditions: typical   Library: NangateOpenCellLibrary
58 Wire Load Model Mode: top
59
60 Startpoint: b2[6] (input port clocked by MY_CLK)
61 Endpoint: Mid_reg_7/Q_reg[13]
62           (rising edge-triggered flip-flop clocked by MY_CLK)
63 Path Group: MY_CLK
64 Path Type: max
65
66 Des/Clust/Port      Wire Load Model      Library
67 -----
68 fiir_lookahead      5K_hvratio_1_1      NangateOpenCellLibrary
69
70 Point                                     Incr      Path
71 -----
72 clock MY_CLK (rise edge)                  0.00      0.00
73 clock network delay (ideal)                0.00      0.00
74 input external delay                      0.50      0.50 f
75 b2[6] (in)                                  0.00      0.50 f
76 Multi_4/a[6] (multiplier_N13_0)           0.00      0.50 f
77 Multi_4/mult_16/a[6] (multiplier_N13_0_DW_mult_tc_1)
78                                           0.00      0.50 f
79 Multi_4/mult_16/U850/ZN (XNOR2_X1)         0.06      0.56 f
80 Multi_4/mult_16/U851/ZN (OAI22_X1)         0.07      0.64 r
81 Multi_4/mult_16/U247/S (FA_X1)             0.12      0.76 f
82 Multi_4/mult_16/U246/S (FA_X1)             0.13      0.89 r
83 Multi_4/mult_16/U838/ZN (AND2_X1)          0.05      0.94 r
84 Multi_4/mult_16/U642/ZN (AOI21_X1)         0.03      0.97 f

```

```

85 Multi_4/mult_16/U618/ZN (OAI21_X1) 0.07 1.04 r
86 Multi_4/mult_16/U568/ZN (AOI21_X1) 0.04 1.07 f
87 Multi_4/mult_16/U860/ZN (OAI21_X1) 0.05 1.13 r
88 Multi_4/mult_16/U873/ZN (AOI21_X1) 0.04 1.16 f
89 Multi_4/mult_16/U571/ZN (OR2_X1) 0.06 1.22 f
90 Multi_4/mult_16/U572/ZN (NAND2_X1) 0.03 1.25 r
91 Multi_4/mult_16/U584/ZN (AOI21_X1) 0.03 1.28 f
92 Multi_4/mult_16/U593/ZN (OAI21_X1) 0.05 1.33 r
93 Multi_4/mult_16/U575/ZN (AOI21_X1) 0.03 1.36 f
94 Multi_4/mult_16/U954/ZN (OAI21_X1) 0.05 1.41 r
95 Multi_4/mult_16/U583/ZN (AOI21_X1) 0.03 1.44 f
96 Multi_4/mult_16/U988/ZN (OAI21_X1) 0.06 1.49 r
97 Multi_4/mult_16/U548/ZN (NAND2_X1) 0.03 1.53 f
98 Multi_4/mult_16/U549/ZN (NAND3_X1) 0.04 1.57 r
99 Multi_4/mult_16/U551/Z (XOR2_X1) 0.07 1.64 r
100 Multi_4/mult_16/product[25] (multiplier_N13_0_DW_mult_tc_1)
101 0.00 1.64 r
102 Multi_4/o[25] (multiplier_N13_0) 0.00 1.64 r
103 Mid_reg_7/D[13] (generic_reg_N14_1) 0.00 1.64 r
104 Mid_reg_7/Q_reg[13]/D (DFFR_X1) 0.01 1.65 r
105 data arrival time 1.65
106
107 clock MY_CLK (rise edge) 1.76 1.76
108 clock network delay (ideal) 0.00 1.76
109 clock uncertainty -0.07 1.69
110 Mid_reg_7/Q_reg[13]/CK (DFFR_X1) 0.00 1.69 r
111 library setup time -0.04 1.65
112 data required time 1.65
113 -----
114 data required time 1.65
115 data arrival time -1.65
116 -----
117 slack (MET) 0.00
118
119
120 1
121 Information: Updating design information... (UID=85)
122
123 *****
124 Report : timing
125 -path full
126 -delay max
127 -max_paths 1
128 Design : fiir_lookahead
129 Version: S-2021.06-SP4
130 Date : Thu Nov 21 20:56:10 2024
131 *****
132
133 Operating Conditions: typical Library: NangateOpenCellLibrary
134 Wire Load Model Mode: top
135
136 Startpoint: b2[6] (input port clocked by MY_CLK)
137 Endpoint: Mid_reg_7/Q_reg[13]
138 (rising edge-triggered flip-flop clocked by MY_CLK)
139 Path Group: MY_CLK
140 Path Type: max
141
142 Des/Clust/Port Wire Load Model Library
143 -----
144 fiir_lookahead 5K_hvrat1o_1_1 NangateOpenCellLibrary
145
146 Point Incr Path
147 -----
148 clock MY_CLK (rise edge) 0.00 0.00
149 clock network delay (ideal) 0.00 0.00
150 input external delay 0.50 0.50 f
151 b2[6] (in) 0.00 0.50 f
152 Multi_4/a[6] (multiplier_N13_0) 0.00 0.50 f
153 Multi_4/mult_16/a[6] (multiplier_N13_0_DW_mult_tc_1)
154 0.00 0.50 f
155 Multi_4/mult_16/U850/ZN (XNOR2_X1) 0.06 0.56 f
156 Multi_4/mult_16/U851/ZN (OAI22_X1) 0.07 0.64 r
157 Multi_4/mult_16/U247/S (FA_X1) 0.12 0.76 f
158 Multi_4/mult_16/U246/S (FA_X1) 0.13 0.89 r
159 Multi_4/mult_16/U838/ZN (AND2_X1) 0.05 0.94 r
160 Multi_4/mult_16/U642/ZN (AOI21_X1) 0.03 0.97 f
161 Multi_4/mult_16/U618/ZN (OAI21_X1) 0.07 1.04 r
162 Multi_4/mult_16/U568/ZN (AOI21_X1) 0.04 1.07 f
163 Multi_4/mult_16/U860/ZN (OAI21_X1) 0.05 1.13 r
164 Multi_4/mult_16/U873/ZN (AOI21_X1) 0.04 1.16 f
165 Multi_4/mult_16/U571/ZN (OR2_X1) 0.06 1.22 f
166 Multi_4/mult_16/U572/ZN (NAND2_X1) 0.03 1.25 r
167 Multi_4/mult_16/U584/ZN (AOI21_X1) 0.03 1.28 f
168 Multi_4/mult_16/U593/ZN (OAI21_X1) 0.05 1.33 r
169 Multi_4/mult_16/U575/ZN (AOI21_X1) 0.03 1.36 f
170 Multi_4/mult_16/U954/ZN (OAI21_X1) 0.05 1.41 r
171 Multi_4/mult_16/U583/ZN (AOI21_X1) 0.03 1.44 f
172 Multi_4/mult_16/U988/ZN (OAI21_X1) 0.06 1.49 r
173 Multi_4/mult_16/U548/ZN (NAND2_X1) 0.03 1.53 f
174 Multi_4/mult_16/U549/ZN (NAND3_X1) 0.04 1.57 r
175 Multi_4/mult_16/U551/Z (XOR2_X1) 0.07 1.64 r
176 Multi_4/mult_16/product[25] (multiplier_N13_0_DW_mult_tc_1)
177 0.00 1.64 r
178 Multi_4/o[25] (multiplier_N13_0) 0.00 1.64 r
179 Mid_reg_7/D[13] (generic_reg_N14_1) 0.00 1.64 r

```

```

180 Mid_reg_7/Q_reg[13]/D (DFFR_X1)          0.01      1.65 r
181 data arrival time                          1.65
182
183 clock MY_CLK (rise edge)                   1.76      1.76
184 clock network delay (ideal)                 0.00      1.76
185 clock uncertainty                          -0.07      1.69
186 Mid_reg_7/Q_reg[13]/CK (DFFR_X1)          0.00      1.69 r
187 library setup time                        -0.04      1.65
188 data required time                          1.65
189 -----
190 data required time                          1.65
191 data arrival time                          -1.65
192 -----
193 slack (MET)                                0.00
194
195
196 1

```

4.24 fiir-lookahead.vhd

```
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3 use IEEE.NUMERIC_STD.ALL;
4
5 entity fiir_lookahead is
6     generic(N:integer:=13; SHAMT:integer:=18);
7
8     port(CLK      : in std_logic ;
9          RSTn     : in std_logic ;
10         Din      : in std_logic_vector((N-1) downto 0);
11         VIN      : in std_logic ;
12         b0       : in std_logic_vector((N-1) downto 0);
13         b1       : in std_logic_vector((N-1) downto 0);
14         a1       : in std_logic_vector((N-1) downto 0);
15         b2       : in std_logic_vector((N-1) downto 0);
16         Dout     : out std_logic_vector((N-1) downto 0);
17         VOUT     : out std_logic);
18
19 end fiir_lookahead;
20
21 architecture behavioral of fiir_lookahead is
22
23     component multiplier
24     generic(N:integer);
25     port(
26         a: in std_logic_vector(N-1 downto 0);
27         b: in std_logic_vector(N downto 0);
28         o: out std_logic_vector(2*N downto 0));
29     end component;
30
31     component adder
32     generic(N:integer);
33     port(
34         a: in std_logic_vector(N downto 0);
35         b: in std_logic_vector(N downto 0);
36         o: out std_logic_vector(N downto 0));
37     end component;
38
39     component DFF_syn
40     port (
41         Din,CK,RSTn : in std_logic ;
42         Q           : out std_logic );
43     end component;
44
45     component generic_reg
46     generic(N:integer);
47     Port (
48         CLK      : in  STD_LOGIC;
49         rst_n    : in  STD_LOGIC;
50         en       : in  STD_LOGIC;
51         D        : in  STD_LOGIC_VECTOR(N-1 downto 0);
52         Q        : out STD_LOGIC_VECTOR(N-1 downto 0));
53     end component;
54
55 --signal
56
57     signal Reg_input_o, Reg_output_D :std_logic_vector((N-1) downto 0);
58     signal Adder1_a, Adder1_o, Adder2_o, Adder3_o: std_logic_vector(N downto 0);
59     signal Multi_1_o_14, Multi_2_o_14, Multi_3_o_14, Multi_4_o_14: std_logic_vector(N downto 0);
60     signal Multi_1_o, Multi_2_o, Multi_3_o, Multi_4_o: std_logic_vector(2*N downto 0);
61     signal FF2_input, FF3_input, FF4_input: std_logic;
62     signal mid_reg_1_Q, mid_reg_2_Q, mid_reg_3_Q, mid_reg_4_Q, mid_reg_5_Q, mid_reg_6_Q, mid_reg_7_Q, mid_reg_8_Q, mid_reg_9_Q: std_logic_vector(N-1 downto 0);
63
64
65
66 begin
67
68     Reg_input: generic_reg
69     generic map (N=>N)
70     port map(
71         CLK=>CLK,
72         rst_n=>RSTn,
73         en=>VIN,
74         D=>Din,
75         Q=>Reg_input_o);
76
77
78     Adder1: adder
79     generic map (N=>N)
80     port map(
81         a=>Adder1_a,
82         b=>mid_reg_5_Q,
83         o=>Adder1_o);
```

```

85
86 Mid_reg_1: generic_reg
87   generic map (N=>(N+1))
88   port map (
89     CLK=>CLK,
90     rst_n=>RSTn,
91     en=> FF2_input,
92     D=> Adder1_o,
93     Q=>Mid_reg_1_Q);
94
95 Mid_reg_2: generic_reg
96   generic map (N=>(N+1))
97   port map (
98     CLK=>CLK,
99     rst_n=>RSTn,
100    en=> FF2_input,
101    D=> Adder1_o,
102    Q=>mid_reg_2_Q);
103
104 Mid_reg_3: generic_reg
105   generic map (N=>(N+1))
106   port map (
107     CLK=>CLK,
108     rst_n=>RSTn,
109     en=> FF2_input,
110     D=> Adder1_o,
111     Q=>mid_reg_3_Q);
112
113 Mid_reg_4: generic_reg
114   generic map (N=>(N+1))
115   port map (
116     CLK=>CLK,
117     rst_n=>RSTn,
118     en=> FF2_input,
119     D=> mid_reg_3_Q,
120     Q=>mid_reg_4_Q);
121
122 Mid_reg_5: generic_reg
123   generic map (N=>(N+1))
124   port map (
125     CLK=>CLK,
126     rst_n=>RSTn,
127     en=> FF2_input,
128     D=> multi_1_o_14,
129     Q=>mid_reg_5_Q);
130
131 Multi_1: multiplier
132   generic map (N=>N)
133   port map (
134     a=>a1, --13
135     b=>mid_reg_3_Q, --14
136     o=>multi_1_o); --27
137
138 Multi_2: multiplier
139   generic map (N=>N)
140   port map (
141     a=>b0, --13
142     b=>mid_reg_1_Q, --14
143     o=>Multi_2_o); --27
144
145 Multi_3: multiplier
146   generic map (N=>N)
147   port map (
148     a=>b1, --13
149     b=>mid_reg_2_Q, --14
150     o=>Multi_3_o); --27
151
152 Multi_4: multiplier
153   generic map (N=>N)
154   port map (
155     a=>b2, --13
156     b=>mid_reg_4_Q, --14
157     o=>Multi_4_o); --27
158
159 Mid_reg_9: generic_reg
160   generic map (N=>(N+1))
161   port map (
162     CLK=>CLK,
163     rst_n=>RSTn,
164     en=> FF3_input,
165     D=> multi_2_o_14,
166     Q=>mid_reg_9_Q);
167
168 Mid_reg_6: generic_reg
169   generic map (N=>(N+1))
170   port map (
171     CLK=>CLK,
172     rst_n=>RSTn,
173     en=> FF3_input,
174     D=> multi_3_o_14,
175     Q=>Mid_reg_6_Q);
176
177 Mid_reg_7: generic_reg
178   generic map (N=>(N+1))
179   port map (

```

```

180 CLK=>CLK,
181 rst_n=>RSTn,
182 en=> FF3_input,
183 D=> multi_4_o_14,
184 Q=>Mid_reg_7_Q);
185
186 Adder2: adder
187   generic map (N=>N)
188   port map (
189     a=>Mid_reg_9_Q,
190     b=>Mid_reg_8_Q,
191     o=>Adder2_o);
192
193 Adder3: adder
194   generic map (N=>N)
195   port map (
196     a=>Mid_reg_6_Q,
197     b=>Mid_reg_7_Q,
198     o=>Adder3_o);
199
200 Mid_reg_8: generic_reg
201   generic map (N=>(N+1))
202   port map (
203     CLK=>CLK,
204     rst_n=>RSTn,
205     en=> FF4_input,
206     D=> Adder3_o,
207     Q=>Mid_reg_8_Q);
208
209
210 Reg_output: generic_reg
211   generic map (N=>N)
212   port map (
213     CLK=>CLK,
214     rst_n=>RSTn,
215     en=>FF4_input,
216     D=>Reg_output_D,
217     Q=>Dout);
218
219 FF1: DFF_syn
220   port map (Din=> VIN,
221     CK=>CLK,
222     RSTn=>RSTn,
223     Q=>FF2_input);
224
225 FF2: DFF_syn
226   port map (Din=> FF2_input,
227     CK=>CLK,
228     RSTn=>RSTn,
229     Q=>FF3_input);
230
231 FF3: DFF_syn
232   port map (Din=> FF3_input,
233     CK=>CLK,
234     RSTn=>RSTn,
235     Q=>FF4_input);
236
237 FF4: DFF_syn
238   port map (Din=> FF4_input,
239     CK=>CLK,
240     RSTn=>RSTn,
241     Q=>vout);
242
243
244 Adder1_a    <= Reg_input_o(N-1)&Reg_input_o;
245 Multi_1_o_14 <= Multi_1_o((2*N-1) downto SHAMT)& "000000"; --number of zeroes =(SHAMT-N)+1=18-13+1=6
246 Multi_2_o_14 <= Multi_2_o((2*N-1) downto SHAMT)& "000000";
247 Multi_3_o_14 <= Multi_3_o((2*N-1) downto SHAMT)& "000000";
248 Multi_4_o_14 <= Multi_4_o((2*N-1) downto SHAMT)& "000000";
249 Reg_output_D <= Adder2_o((N-1) downto 0); -- scale 14bit(result of adder2 to 13 bit(input of reg_output)
250
251
252 end behavioral;

```